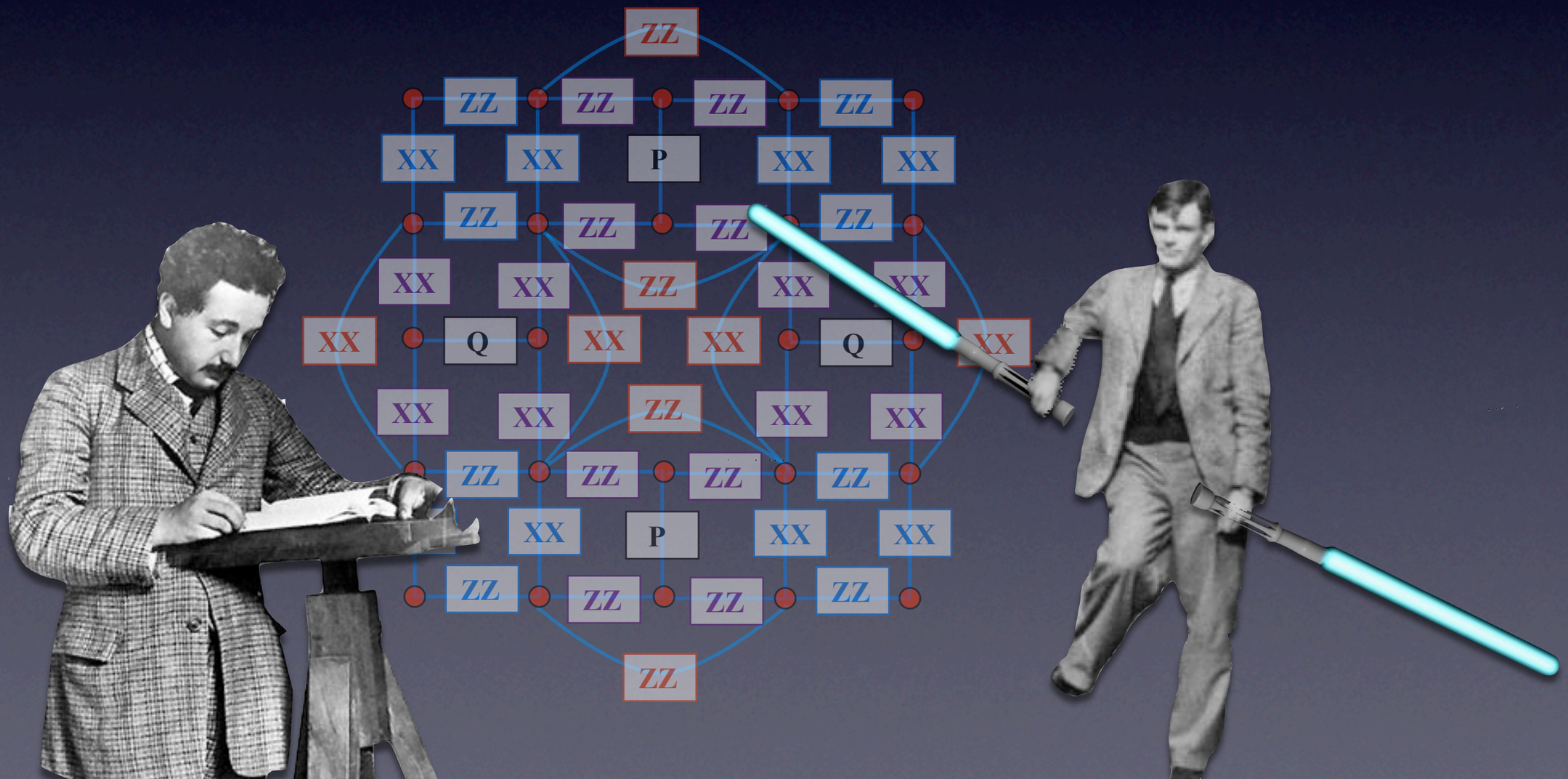
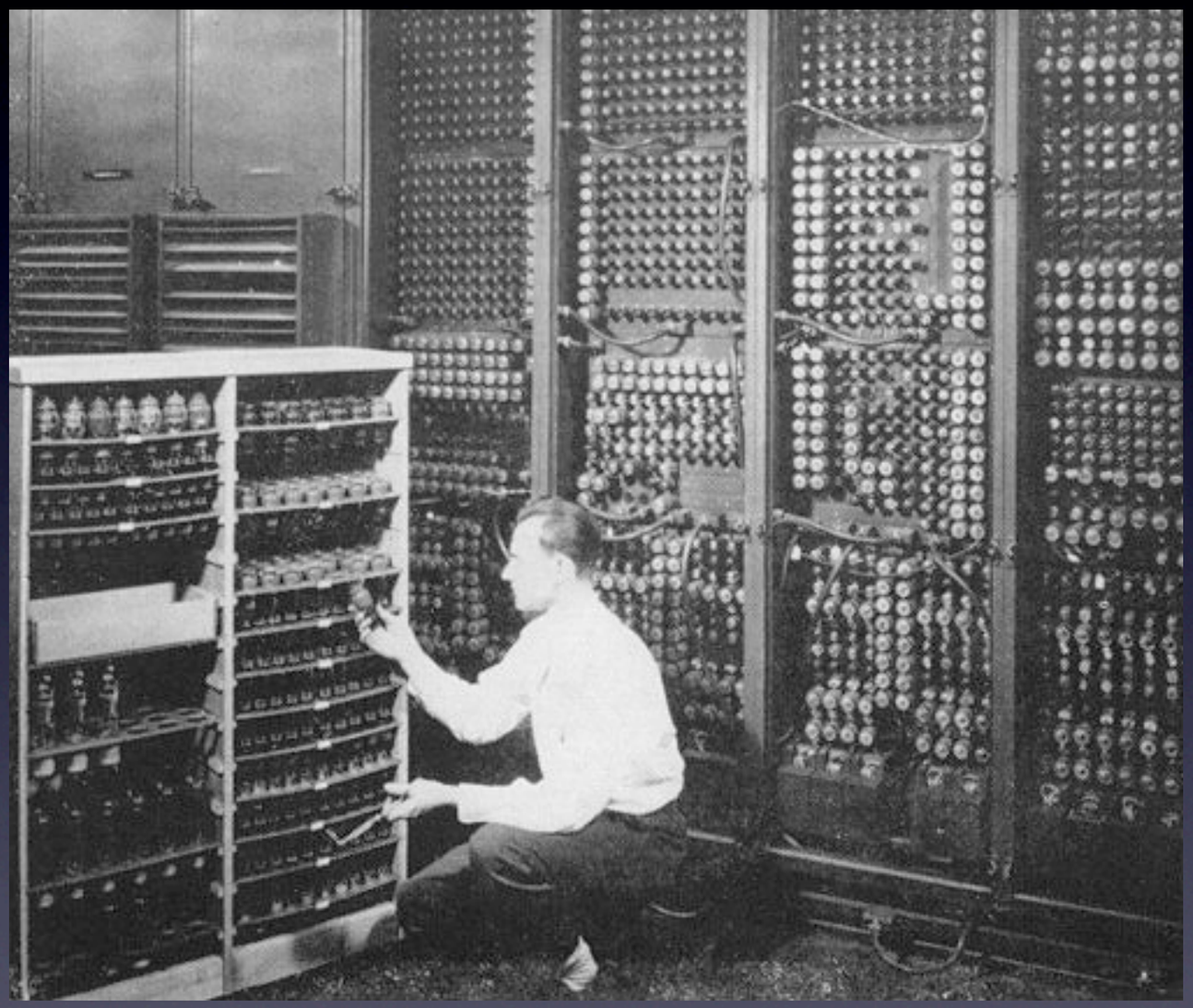
The Race to Build a Quantum Computer

Dave Bacon
University of Washington

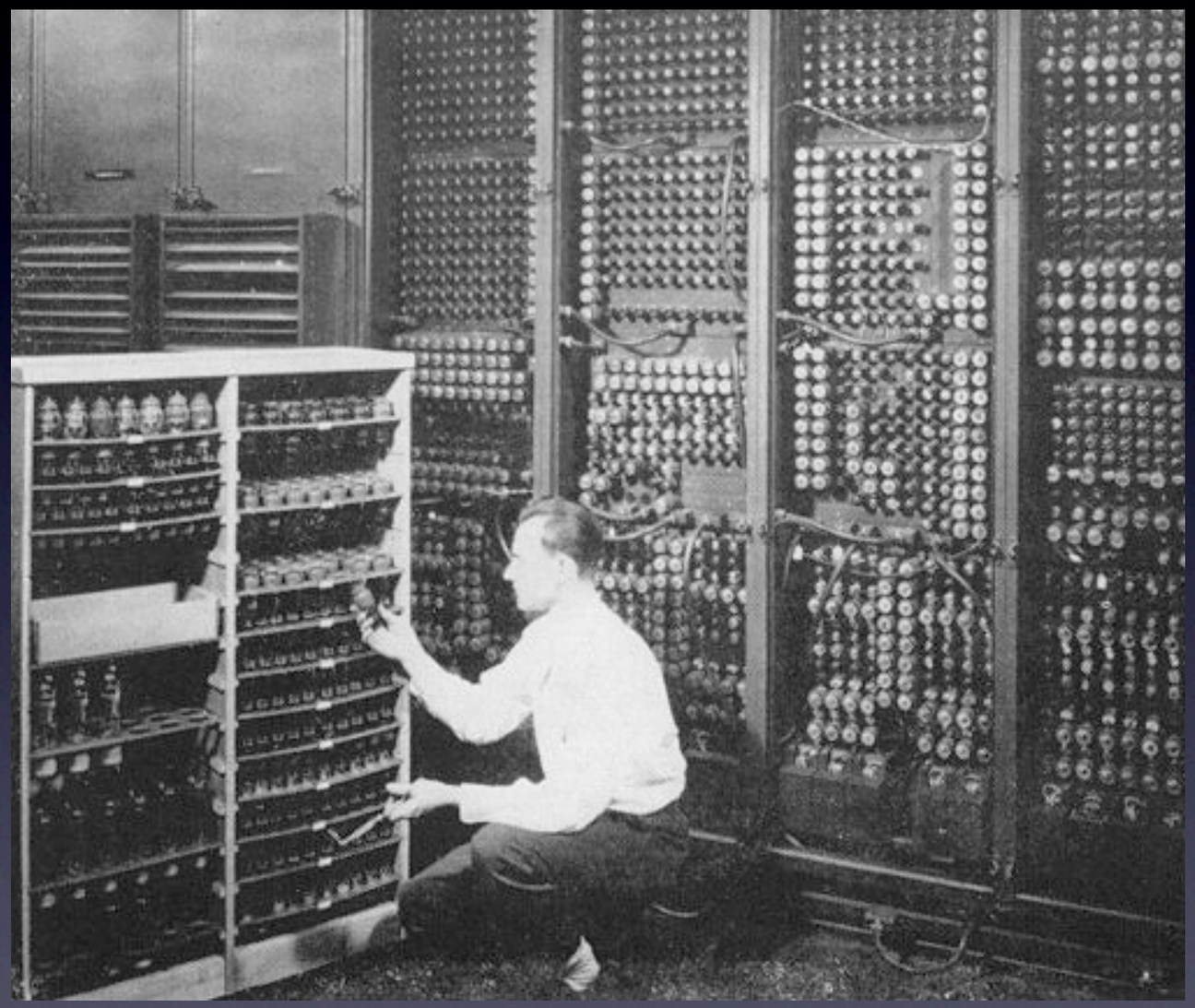


ENIAC, 1946



- 17,468 vacuum tubes
- 7,200 crystal diodes
- 1,500 relays
- 70,000 resistors
- 10,000 capacitors
- ~5,000,000 hand-soldered joints

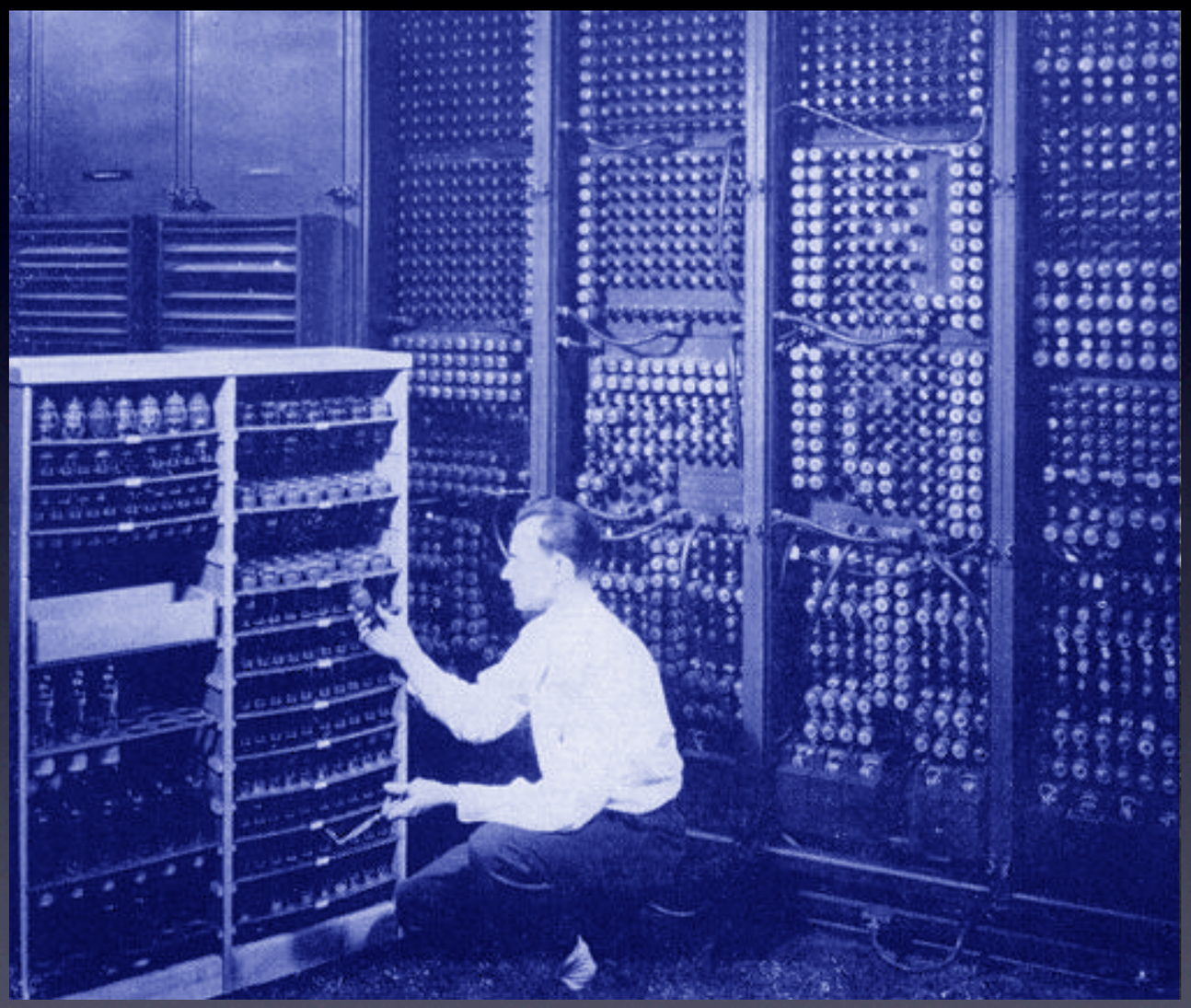
ENIAC, 1946



- 17,468 vacuum tubes
- 7,200 crystal diodes
- 1,500 relays
- 70,000 resistors
- 10,000 capacitors
- ~5,000,000 hand-soldered joints

~ one vacuum tube failure / day
(equivalent to ~ one transistor failure /second today)

ENIAC, 1946



- 17,468 vacuum tubes
- 7,200 crystal diodes
- 1,500 relays
- 70,000 resistors
- 10,000 capacitors
- ~5,000,000 hand-soldered joints

~ one vacuum tube failure / day

(equivalent to ~ one transistor failure /second today)

Alternative History

Imagine: no transistors, no integrated circuits...

Alternative History

Imagine: no transistors, no integrated circuits...



Alternative History

Imagine: no transistors, no integrated circuits...





Alternative History

Imagine: no transistors, no integrated circuits...







Alternative History

Imagine: no transistors, no integrated circuits...



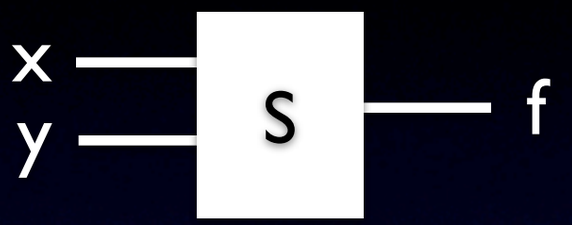




Could we build a computer out of unreliable components?

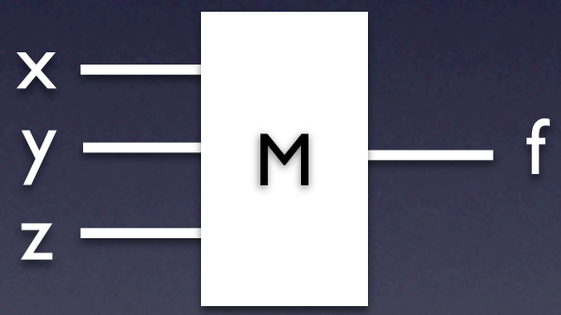
J. von Neumann, "Probabilistic Logics and the Synthesis of Reliable Organism from Unreliable Components" 1956

Von Neumann's Model



NAND gate

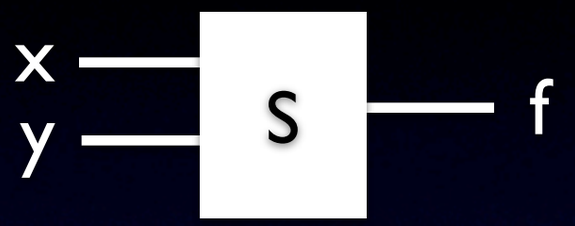
x	y	f
0	0	1
0	1	1
1	0	1
1	1	0



majority gate

x	y	z	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

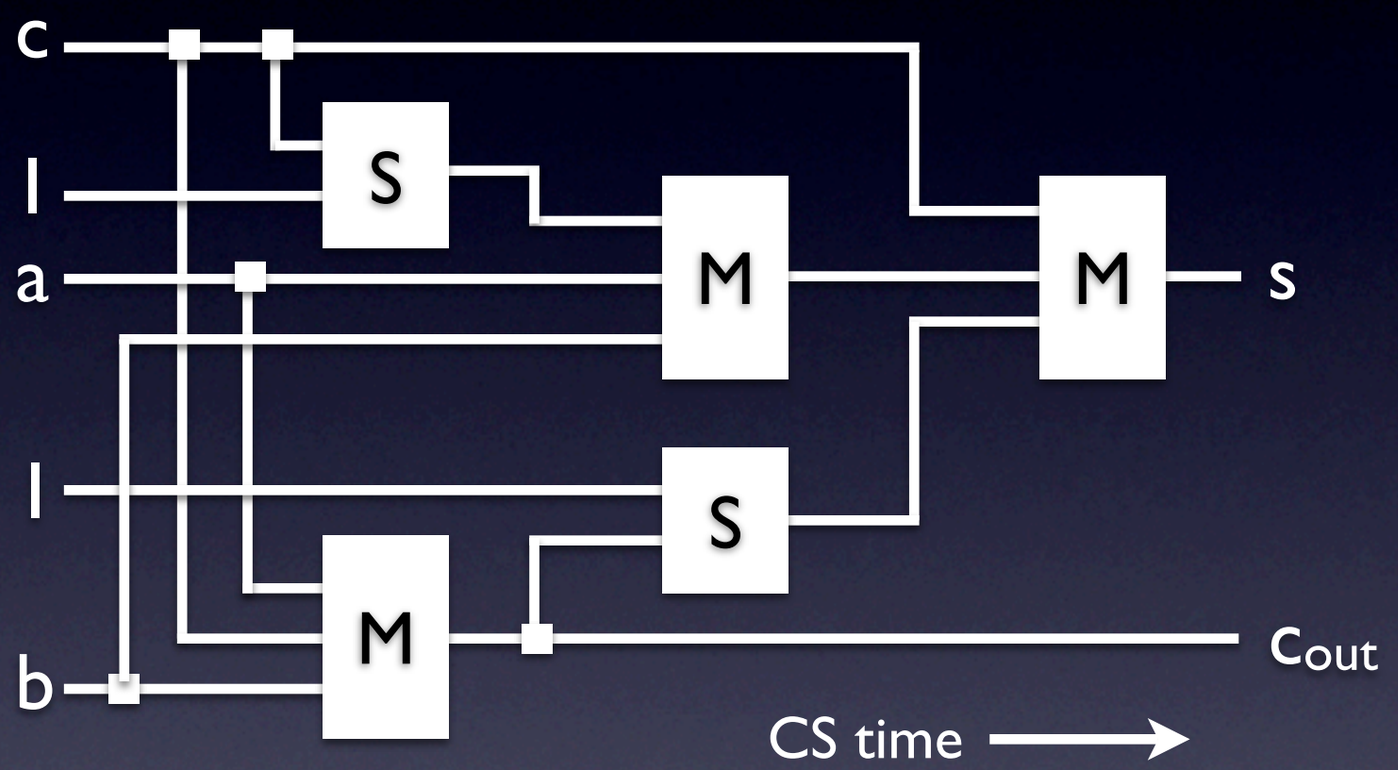
Von Neumann's Model

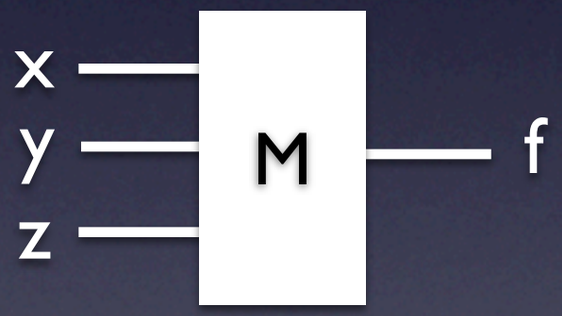


NAND gate

x	y	f
0	0	1
0	1	1
1	0	1
1	1	0

full adder

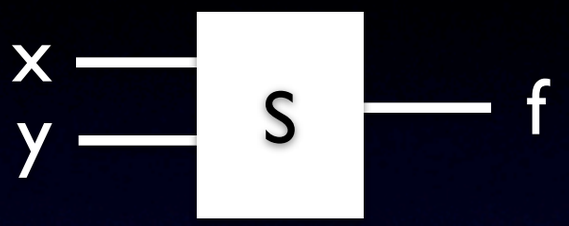




majority gate

x	y	z	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

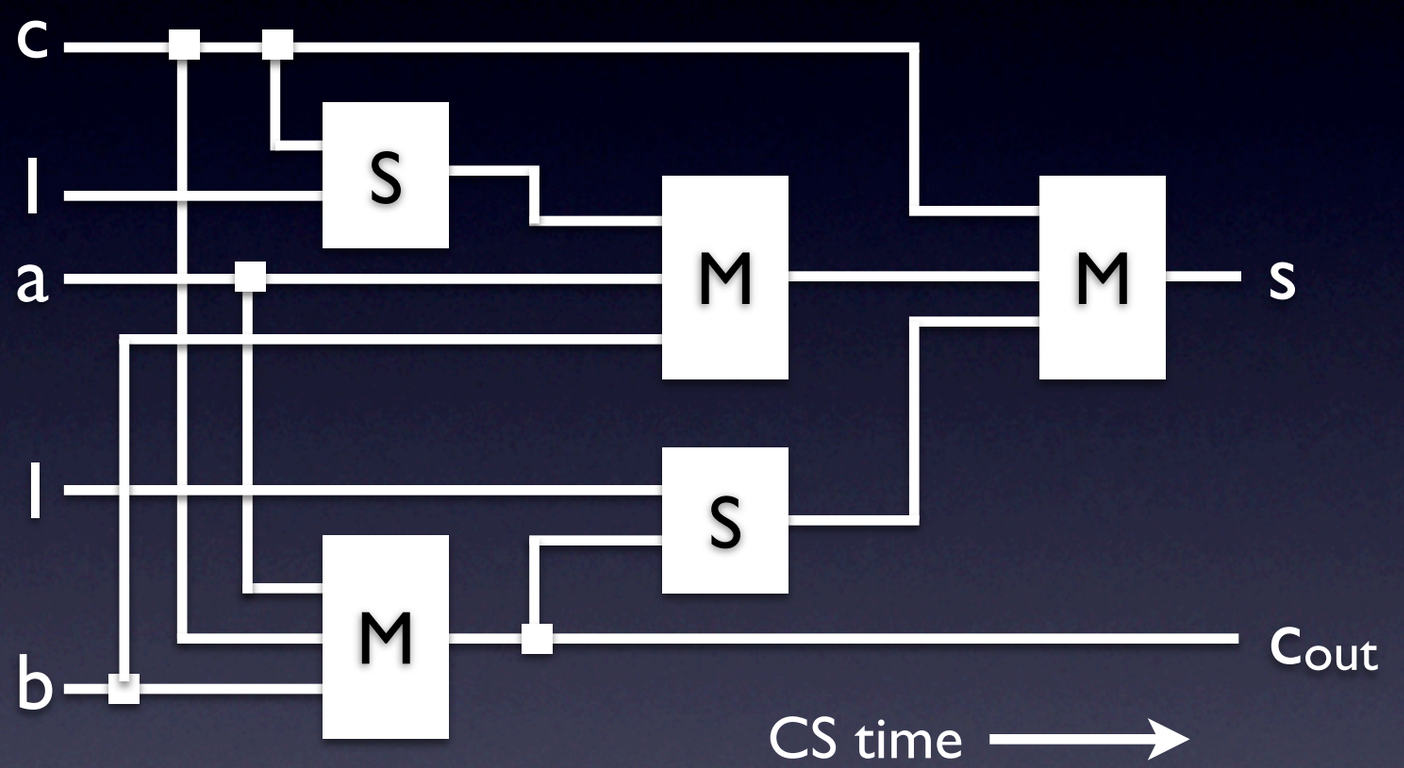
Von Neumann's Model

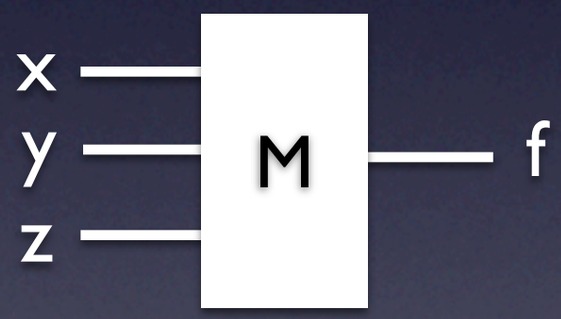


NAND gate

x	y	f
0	0	1
0	1	1
1	0	1
1	1	0

full adder





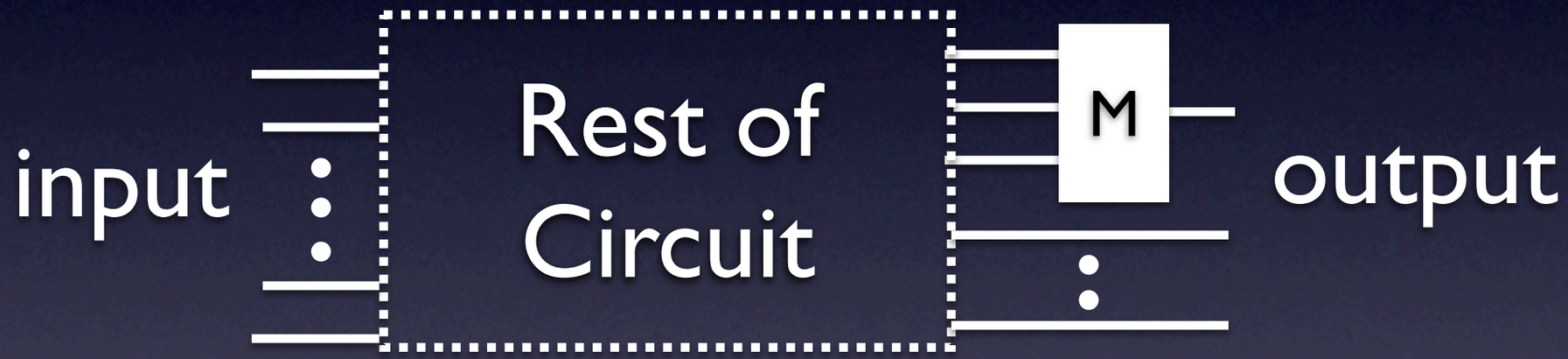
majority gate

x	y	z	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

suppose each gate fails (outputs wrong answer) *independently* and with probability p

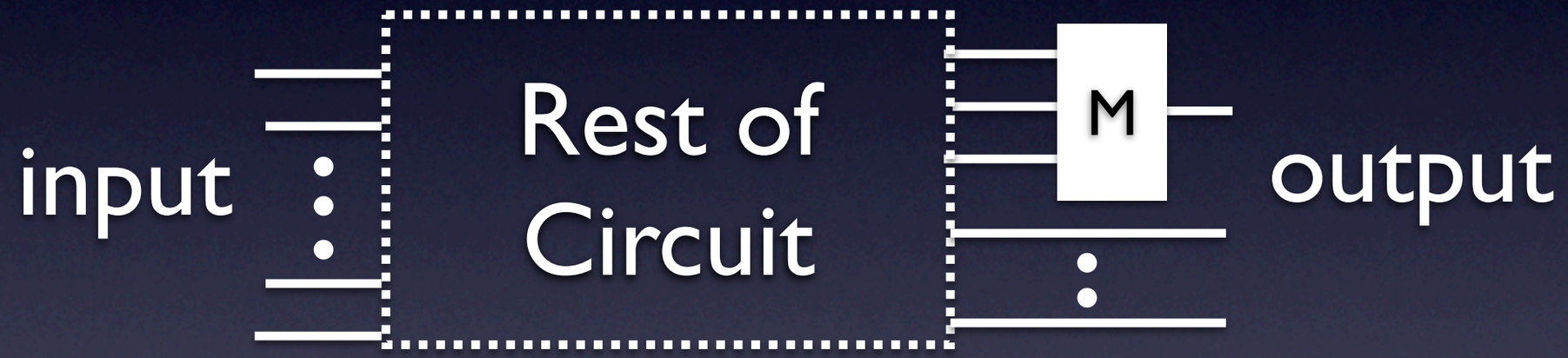
Von Neumann's Conundrum

Observation: If all gates fail with probability p, then an output bit of any circuit most fail with at least probability p

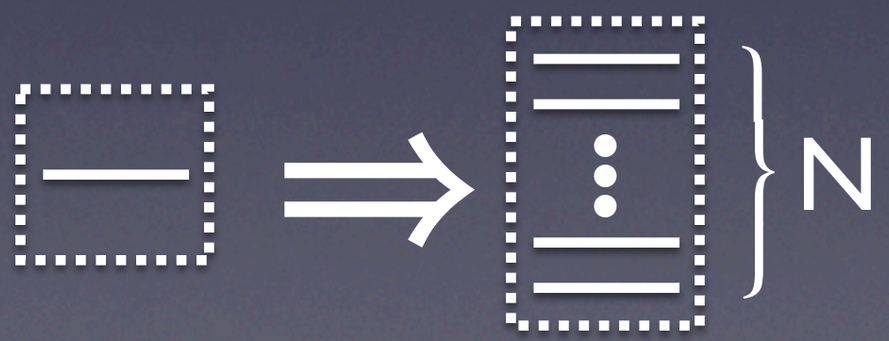


Von Neumann's Conundrum

Observation: If all gates fail with probability p, then an output bit of any circuit most fail with at least probability p



"Loophole": replace wire by "bundle" of N wires



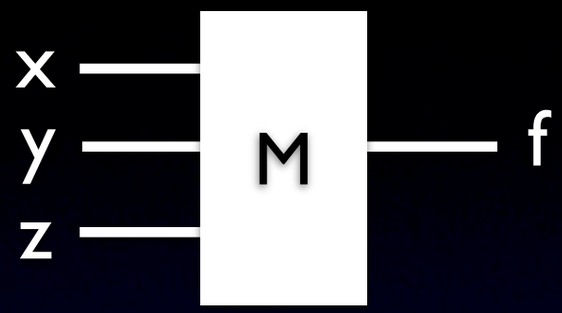
$\leq \beta N$ 0s decode as "1"

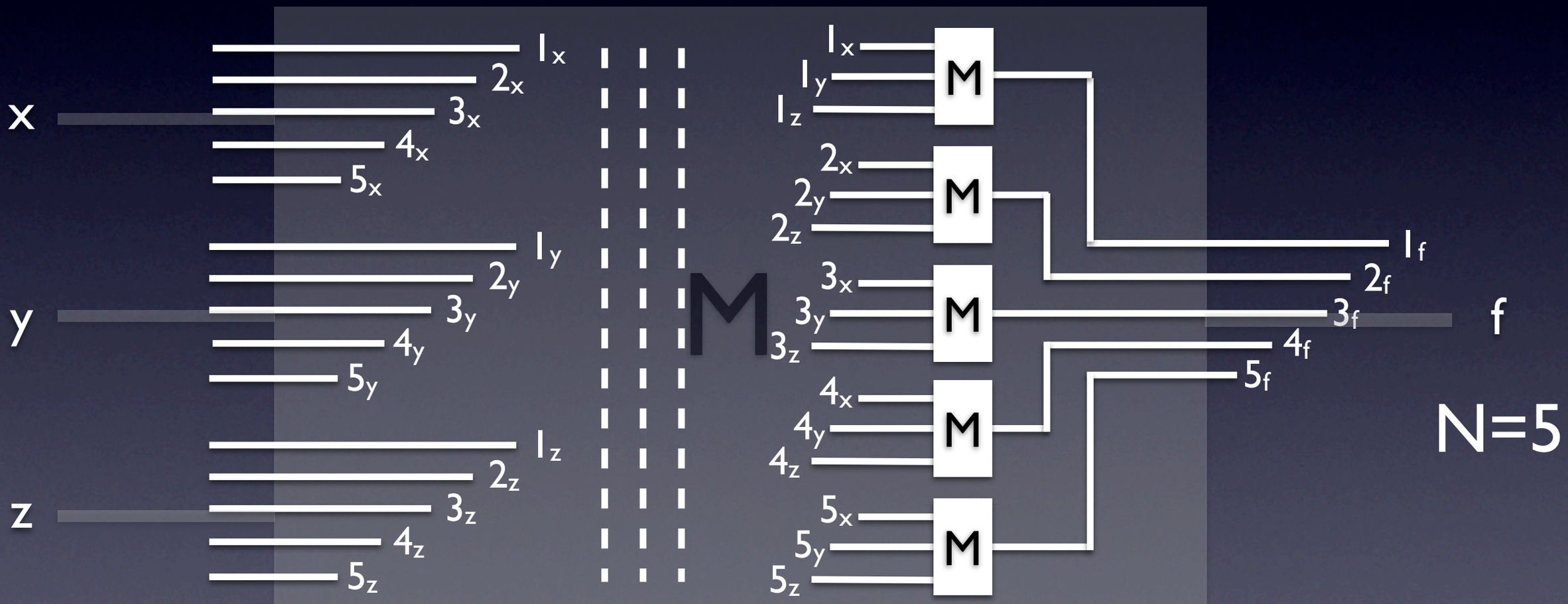
$\geq (1-\beta)N$ 0s decode as "0"

$0<\beta<1/2$

Von Neumann's Solution I

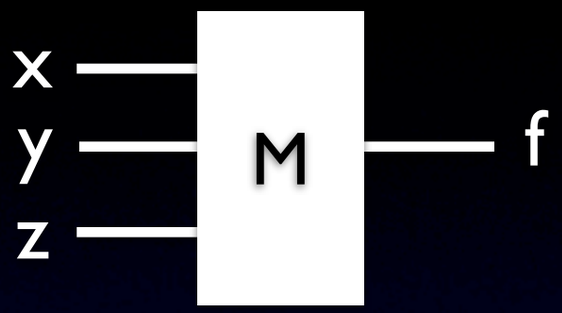
Simulate the majority gate redundantly:

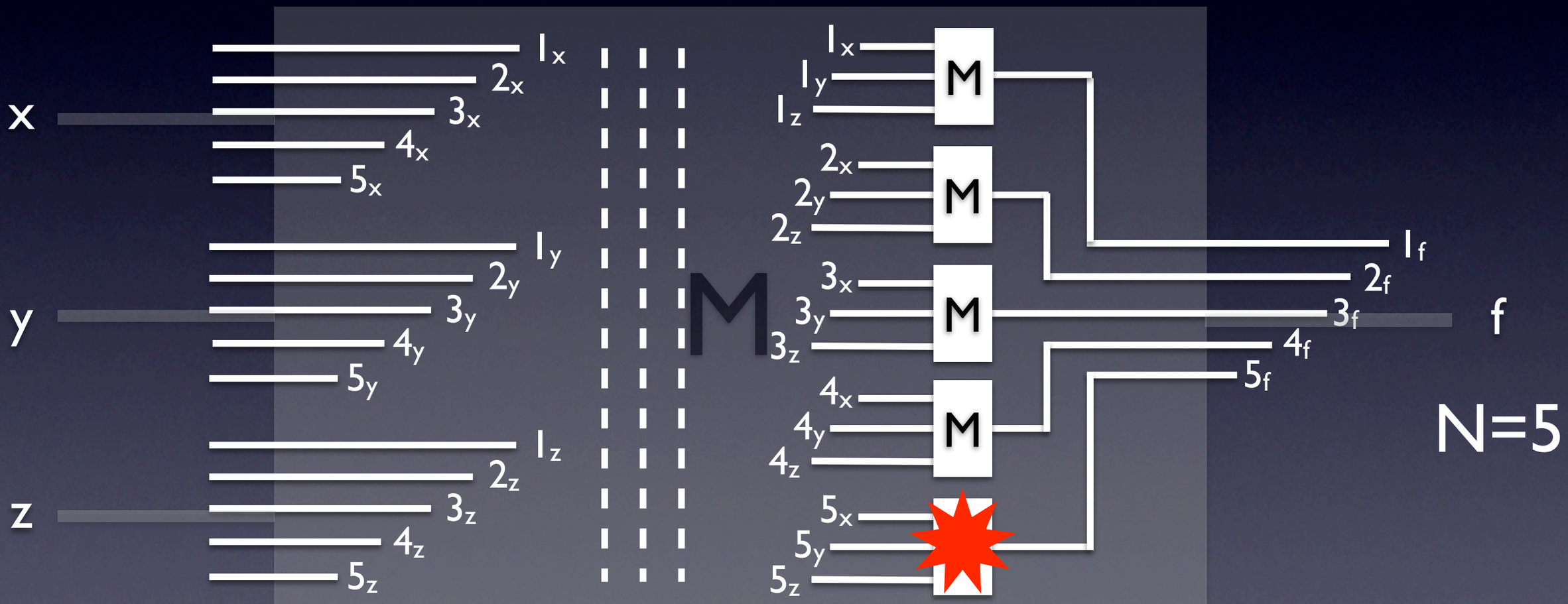




Von Neumann's Solution I

Simulate the majority gate redundantly:

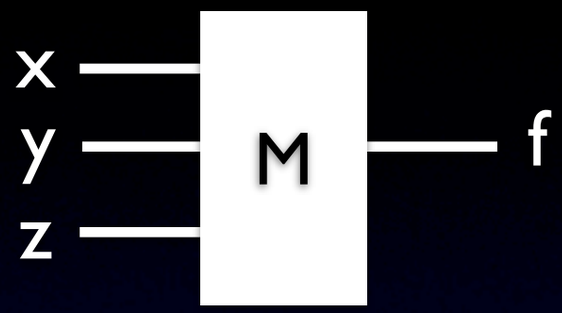


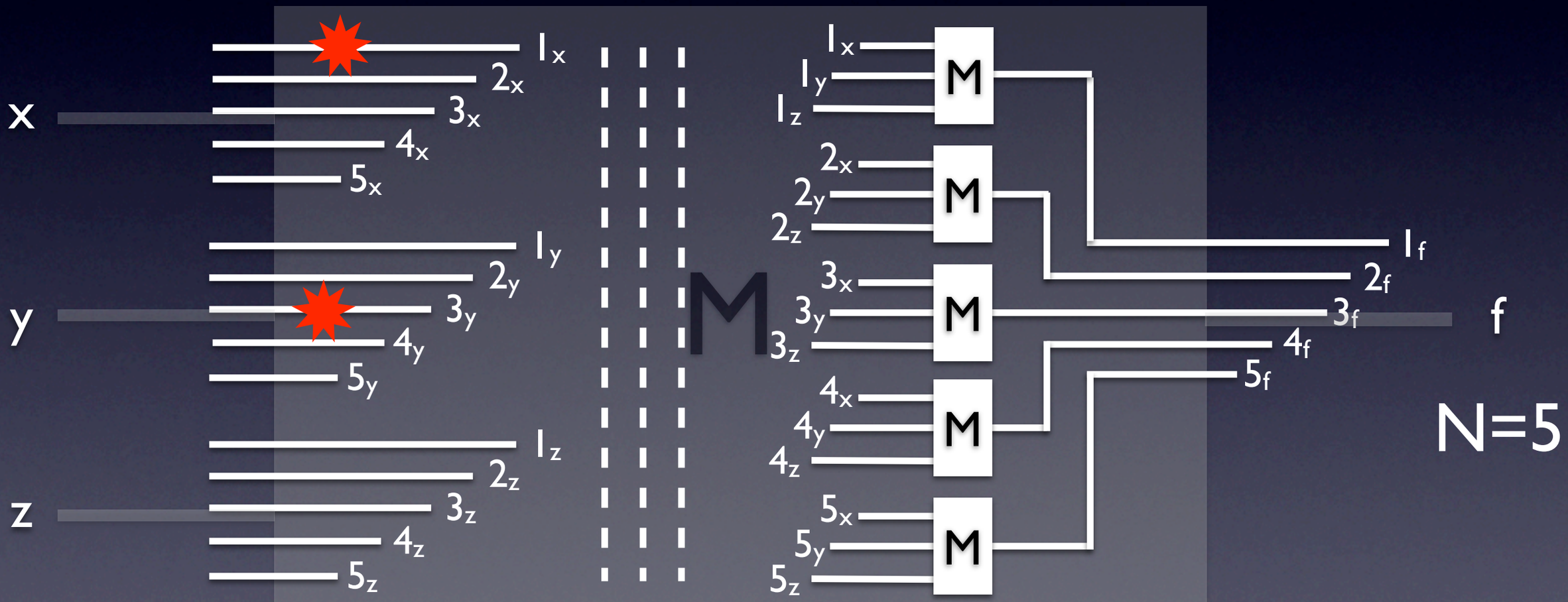


1. M gates may fail

Von Neumann's Solution I

Simulate the majority gate redundantly:



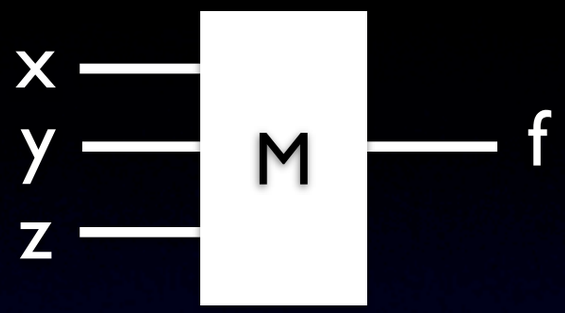


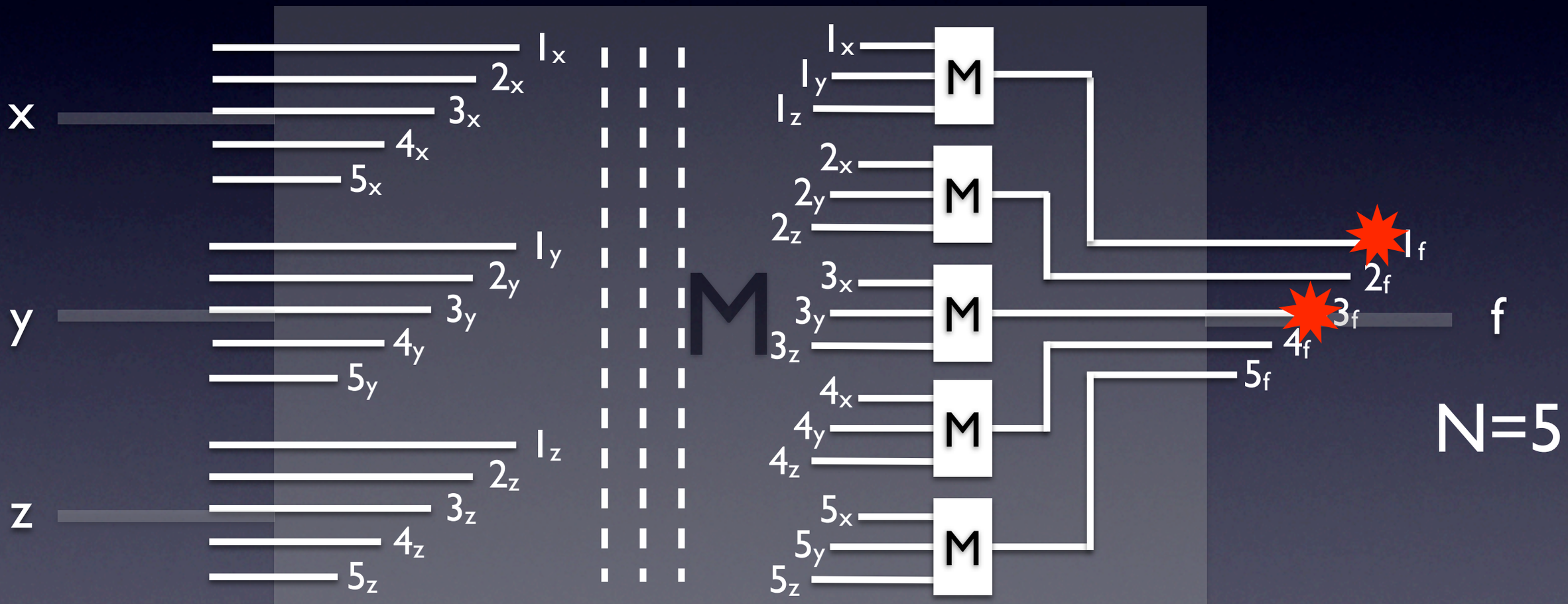
2. incoming errors may add

1. M gates may fail

Von Neumann's Solution I

Simulate the majority gate redundantly:



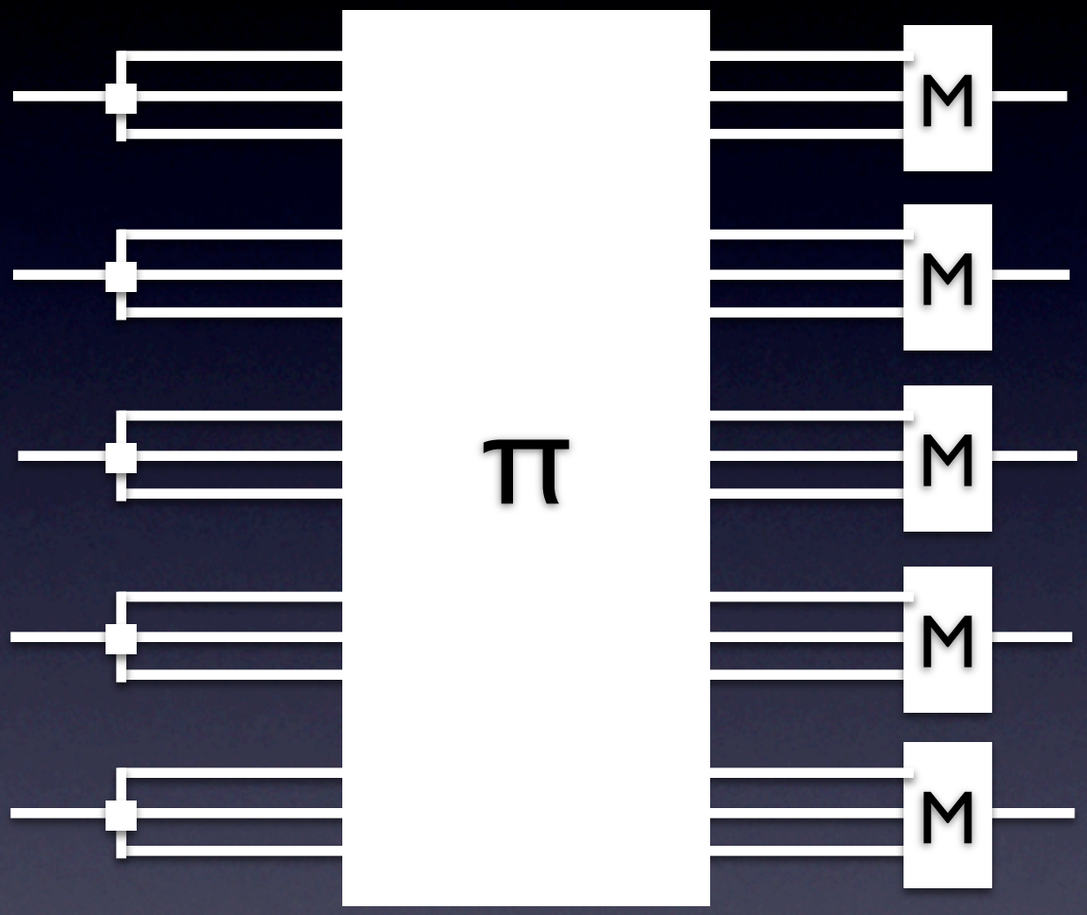


2. incoming errors may add

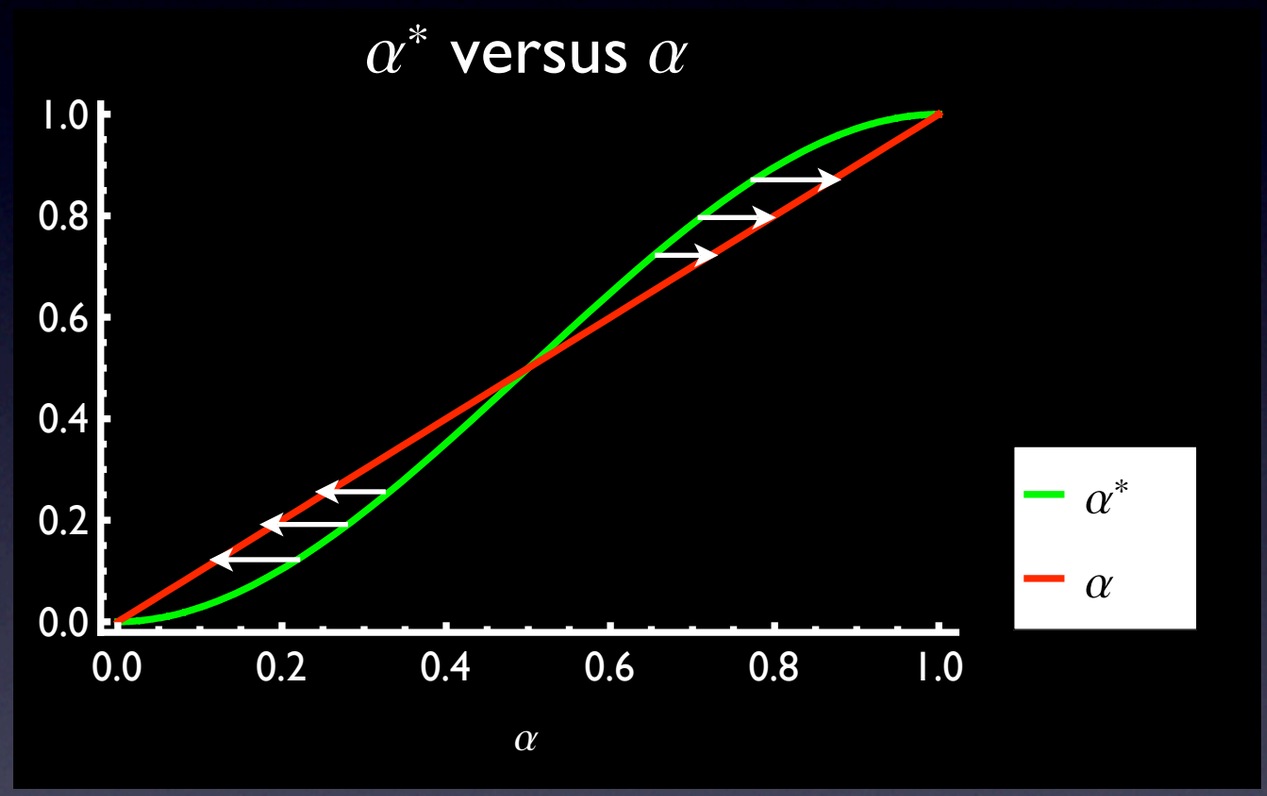
1. M gates may fail

Von Neumann's Solution 2

"Restorative organ":

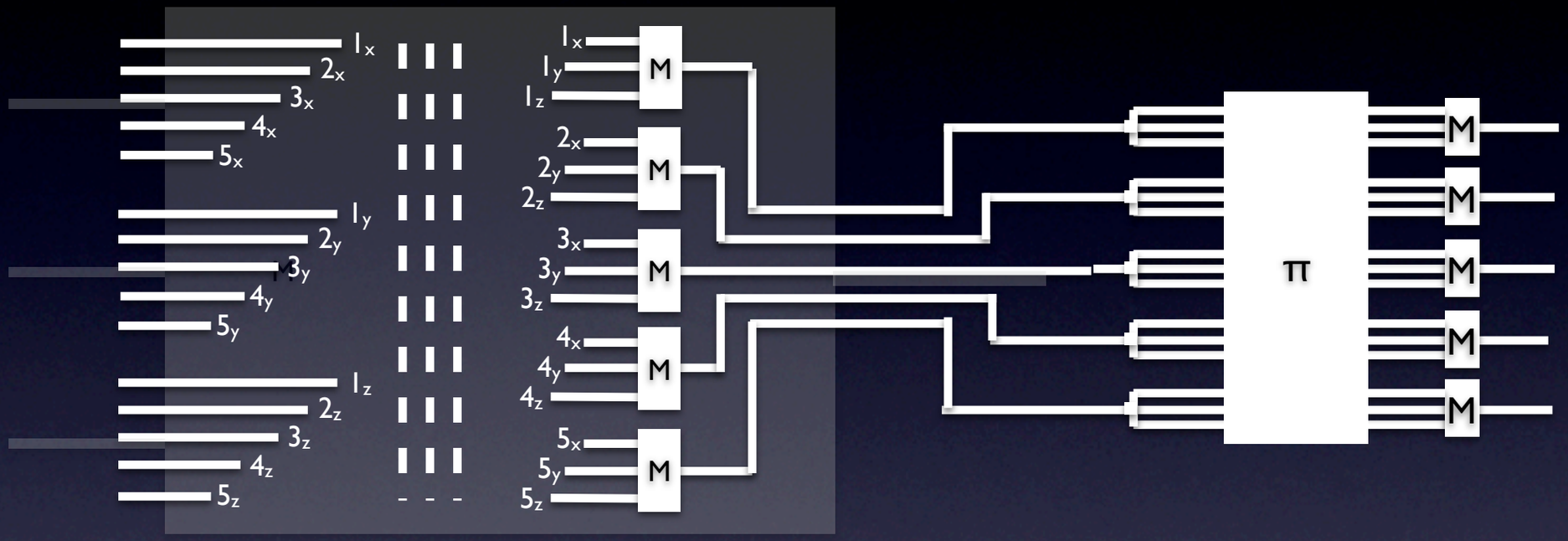


random permutation



αN "0s" $\Rightarrow$ $[3\alpha^2(1-\alpha)+\alpha^3]N$ "0s" = $\alpha^* N$ "0"s

Von Neumann's Solution 1&2



gate error correction

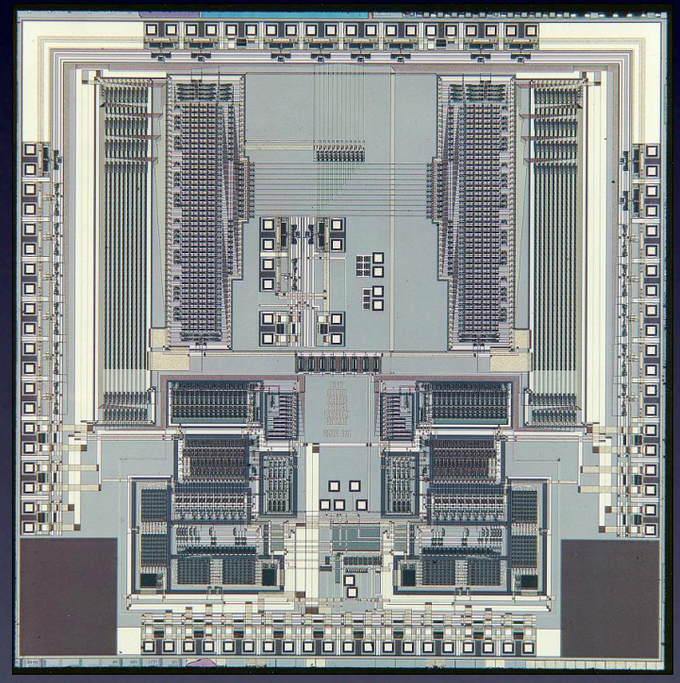
Theorem (von Neumann): A circuit of g gates can be made to fail with probability ε using O(g log (g/ε)) gates which fail with probability p, assuming $p<p_{th}$ (vN estimated $p_{th} \sim 1\%$)



Dave? Did someone forget to tell you this is a physics colloquium?

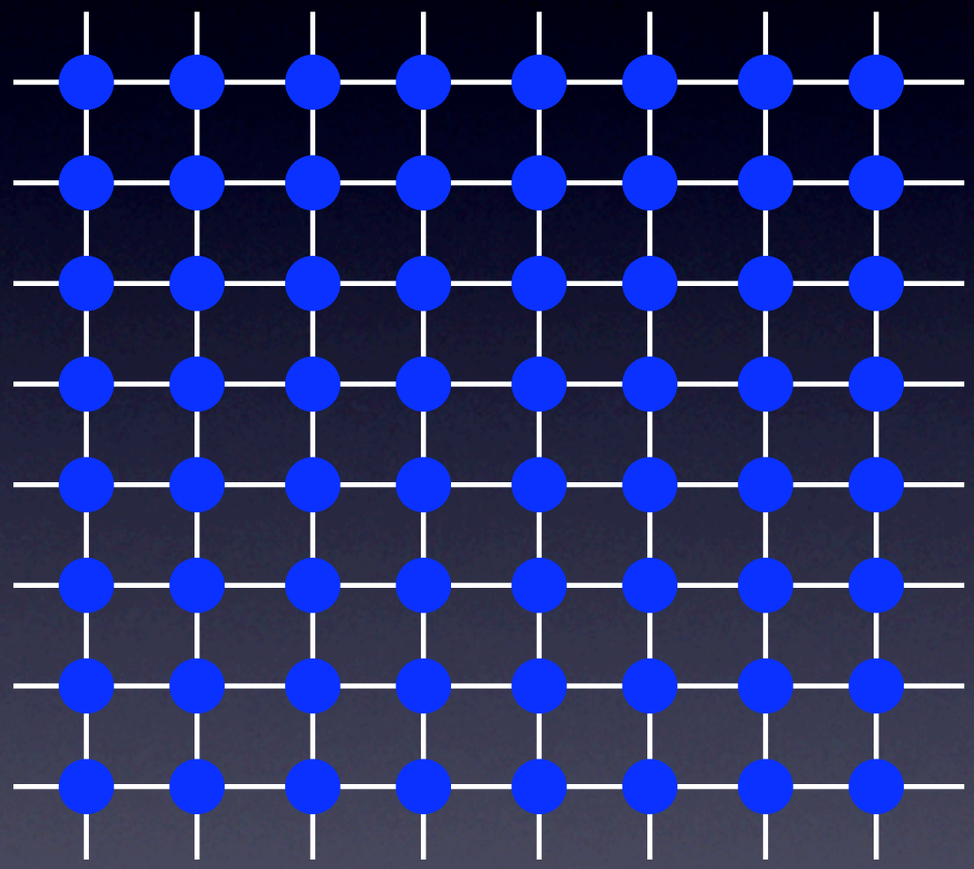
We Don't Need Your Engineering





There are distinct *physical* reasons why robust classical computation is possible

Cartoon Guide to Classical Physical Fault-Tolerance



$+1$

-1

spin

$s_i \in \{+1, -1\}$

Energy

$$E = -J \sum_{\langle i,j \rangle} s_i s_j$$

neighbors

total magnetization

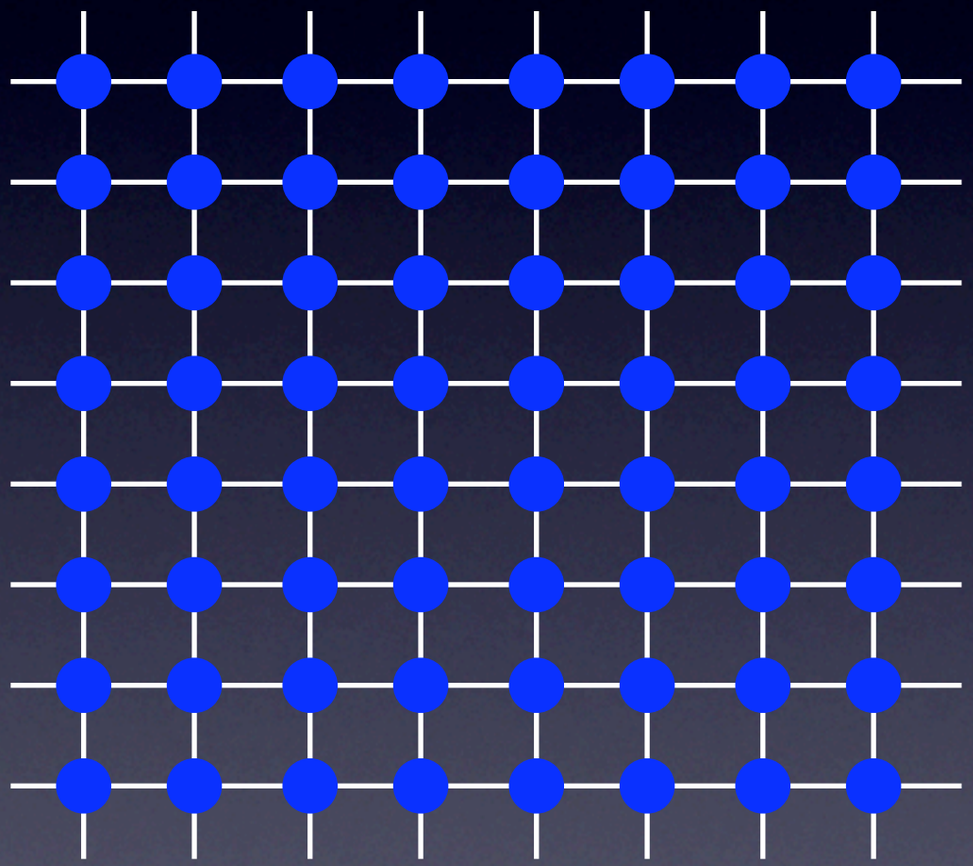
$$M = \sum_i s_i$$

"0" as $M > N_0$

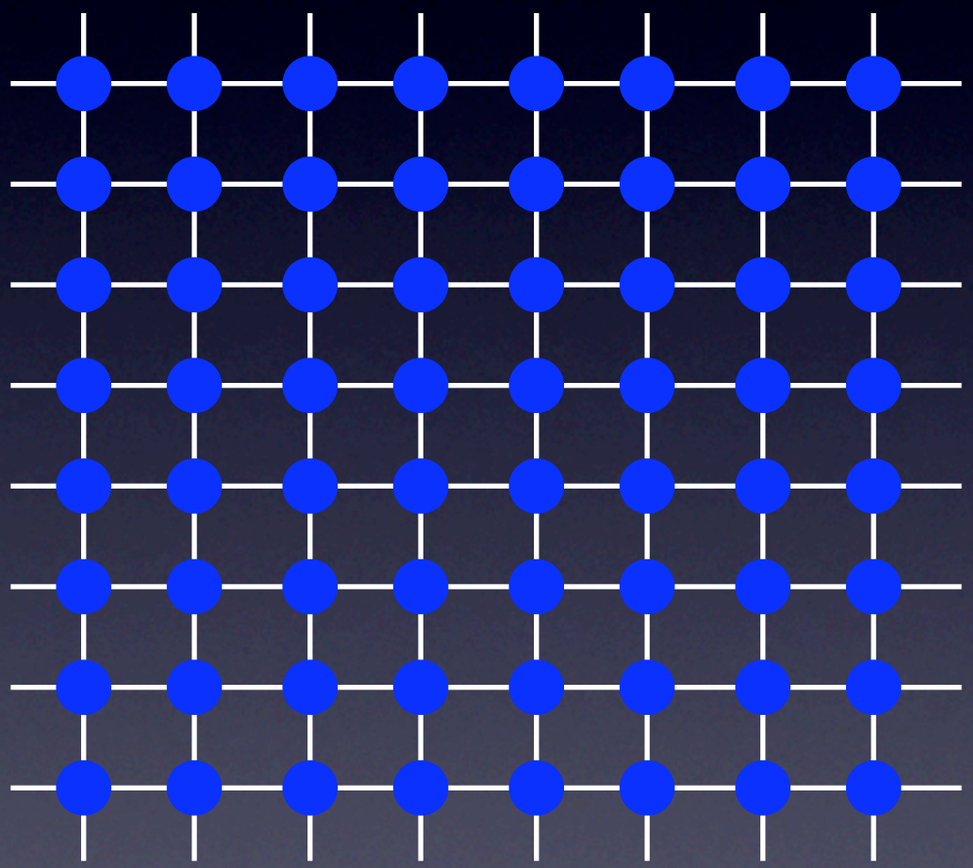
"1" as $M < -N_0$

Cartoon Guide to Classical Physical Fault-Tolerance

Simple noise model



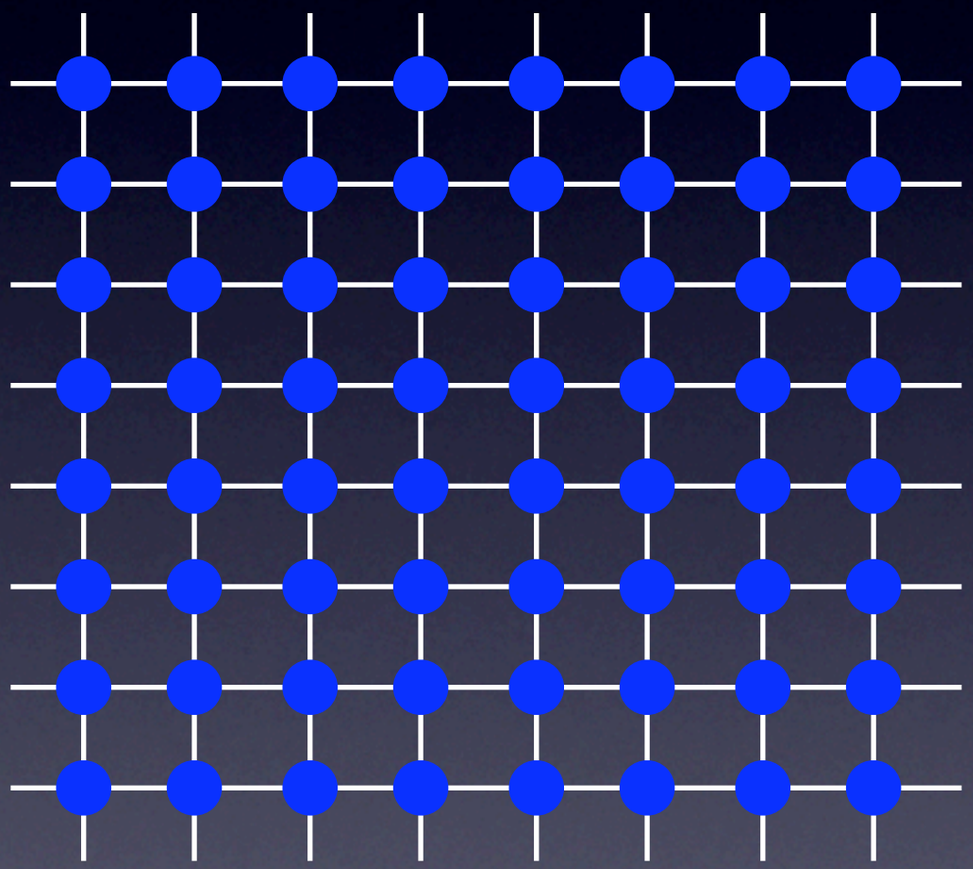
Cartoon Guide to Classical Physical Fault-Tolerance



Simple noise model

1. Pick spin at random

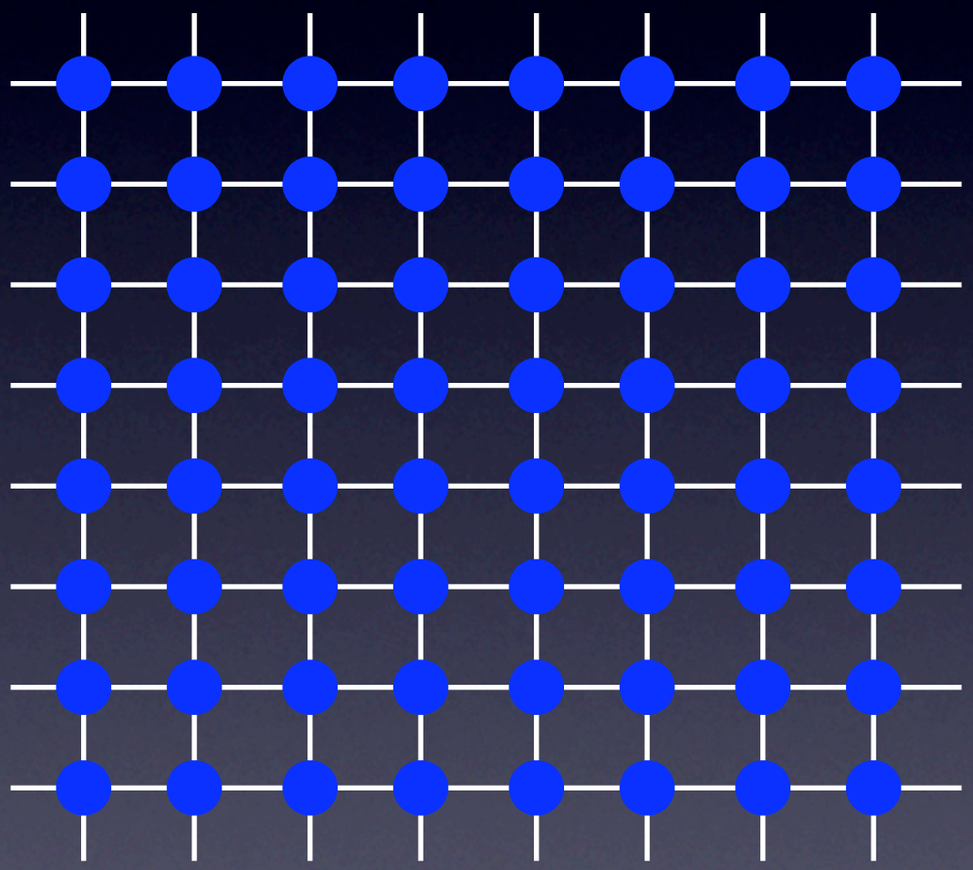
Cartoon Guide to Classical Physical Fault-Tolerance



Simple noise model

1. Pick spin at random
2. If flipping spin would decrease energy
   a. flip spin

Cartoon Guide to Classical Physical Fault-Tolerance



"bare" noise time step

N_r runs of 1-3

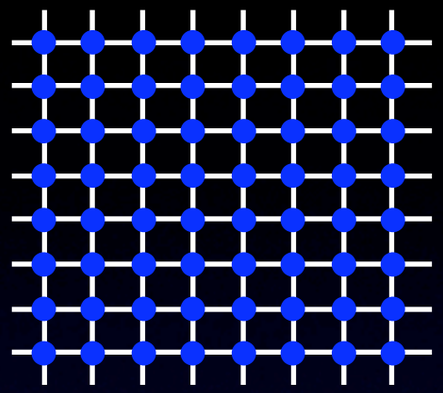
Simple noise model

1. Pick spin at random
2. If flipping spin would decrease energy
   a. flip spin
3. If flipping spin would increase energy:
   a. flip spin with probability

$$\exp(-\beta \Delta E) \quad \beta = T^{-1}$$

Memory

2D

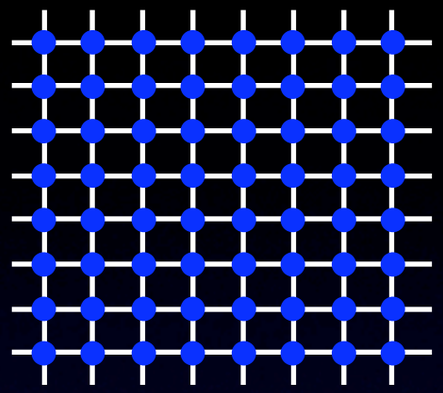


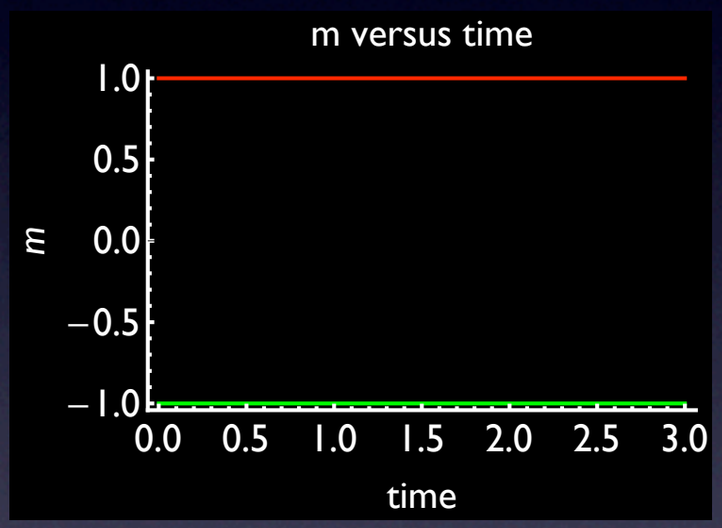
1D



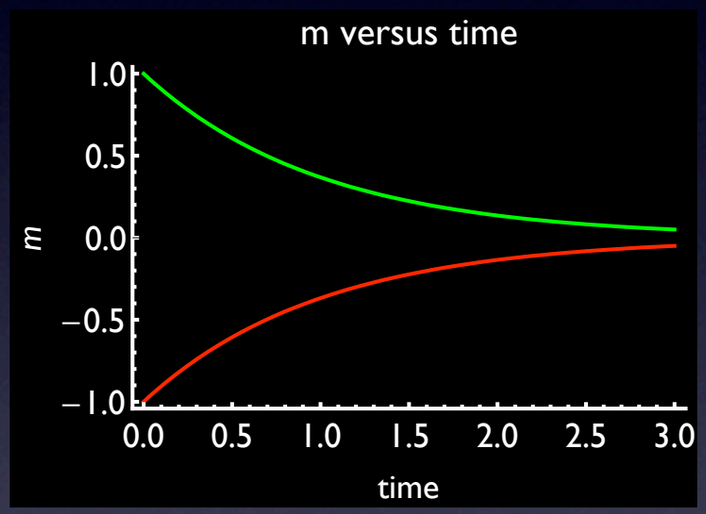
Memory

2D



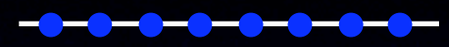


$T < T_C$



$T > T_C$

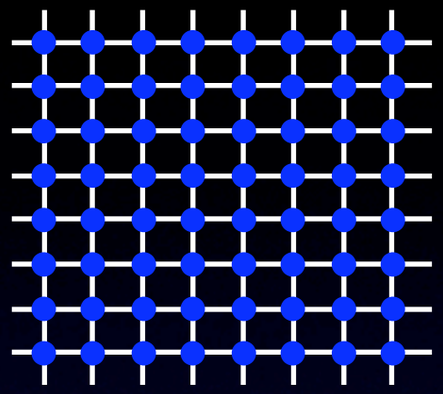
1D

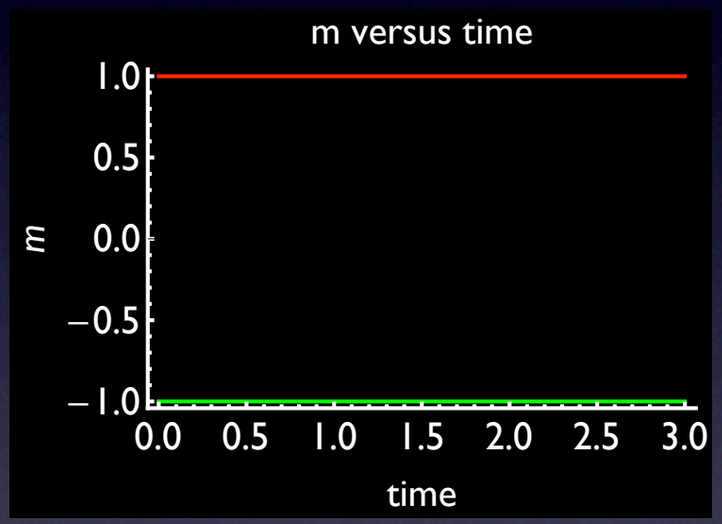


$\beta = T^{-1}$

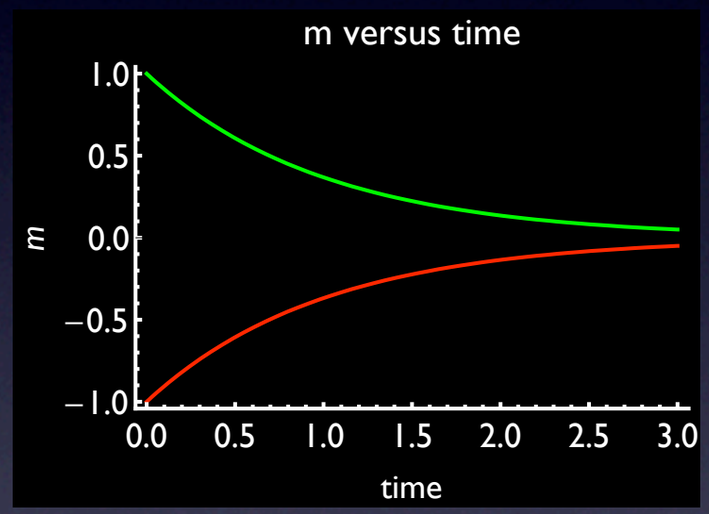
Memory

2D





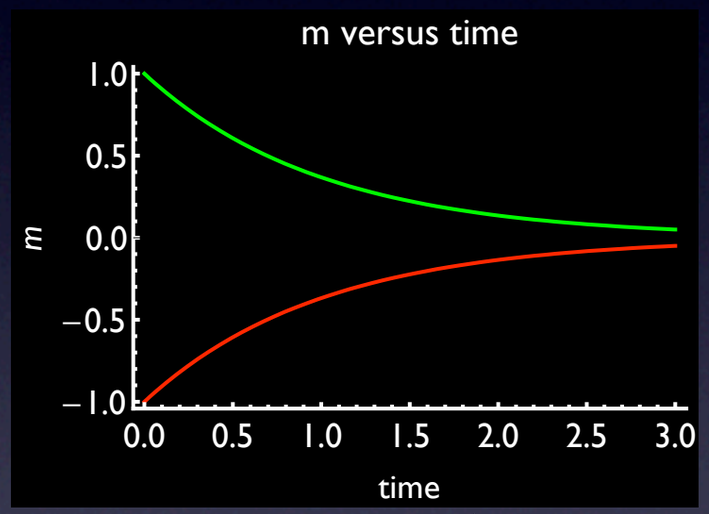
$T < T_C$



$T > T_C$

1D



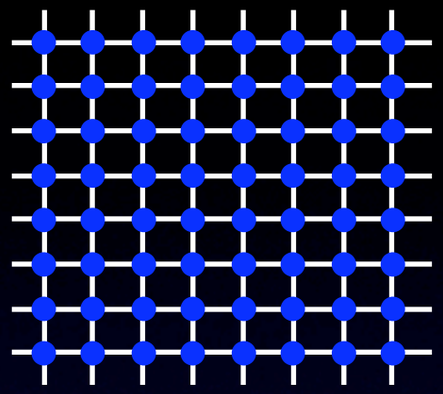


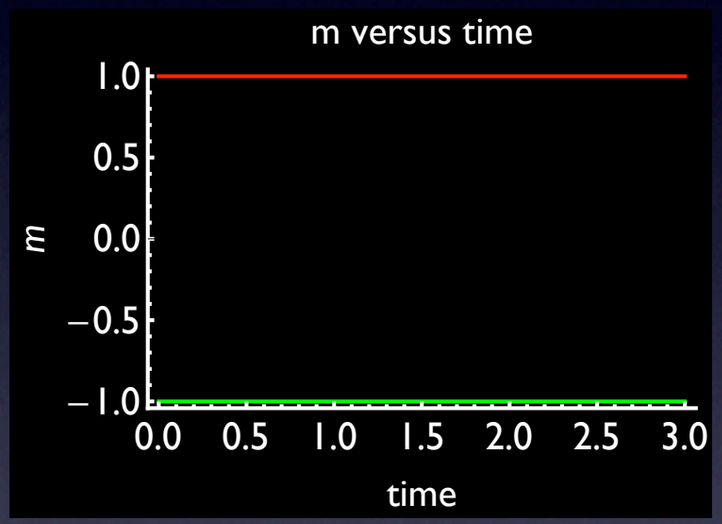
rate ~ $\exp(-c\beta J)$

Memory

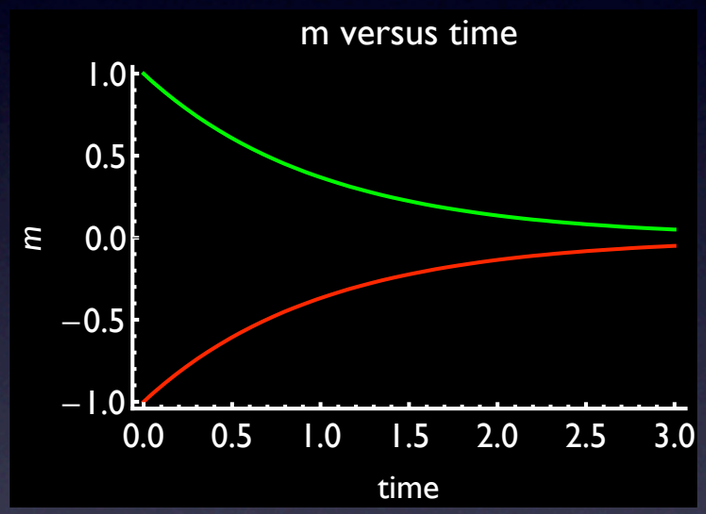
$\beta = T^{-1}$

2D

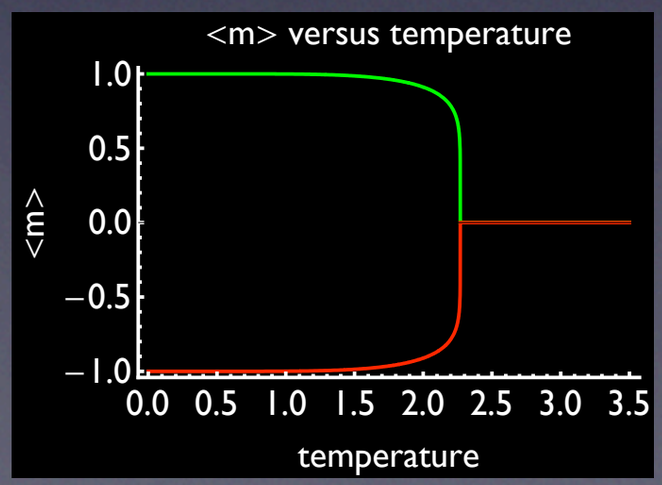




$T < T_C$

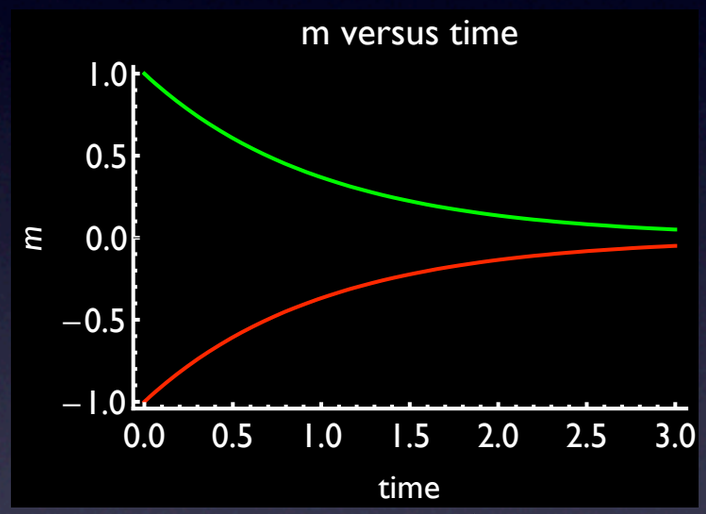


$T > T_C$

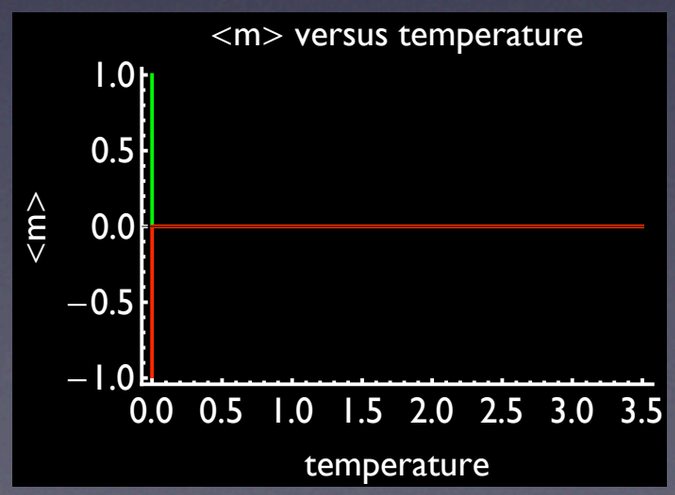


1D



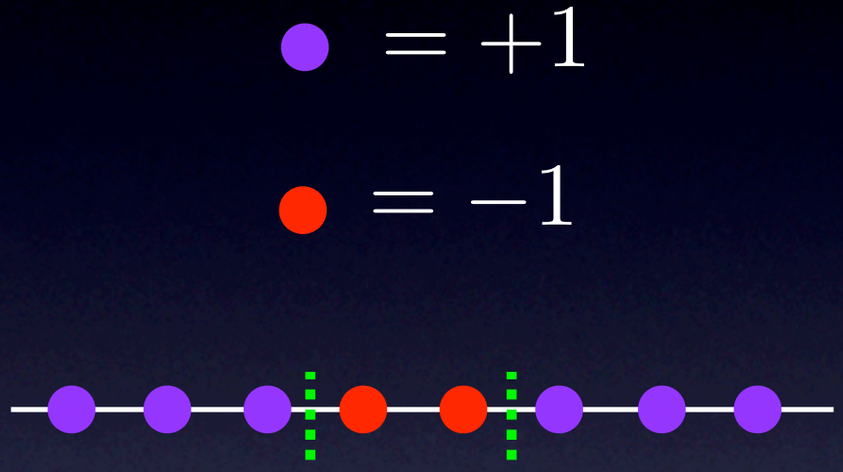


rate ~ $\exp(-c\beta J)$

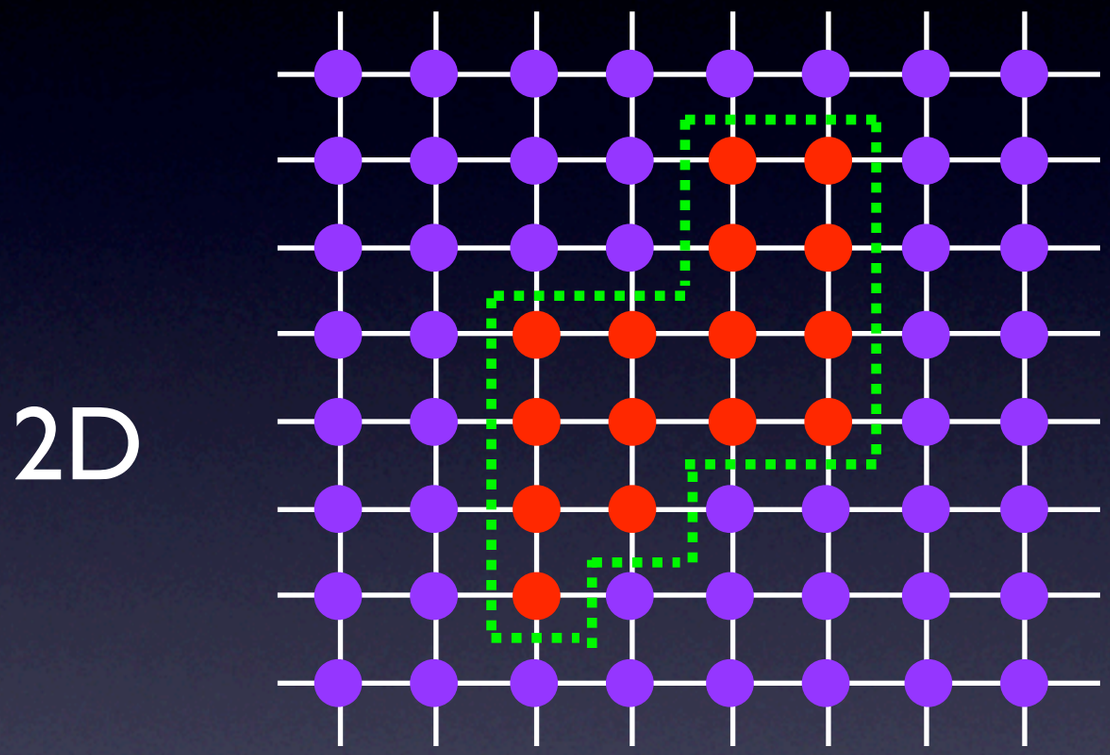


Self-Correcting

Calculate: free energy change to flip all spins



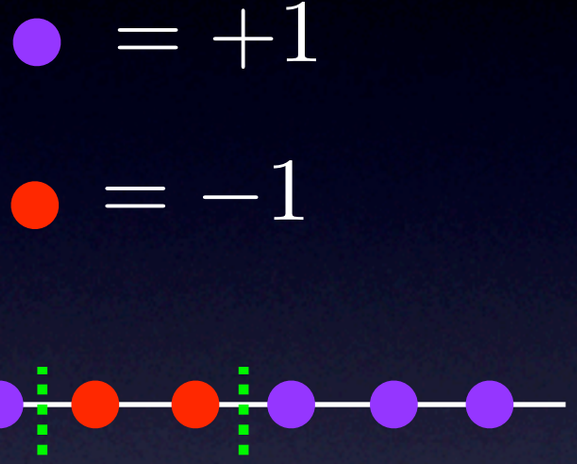
$E \propto$ Constant



$E \propto$ Perimeter

Self-Correcting

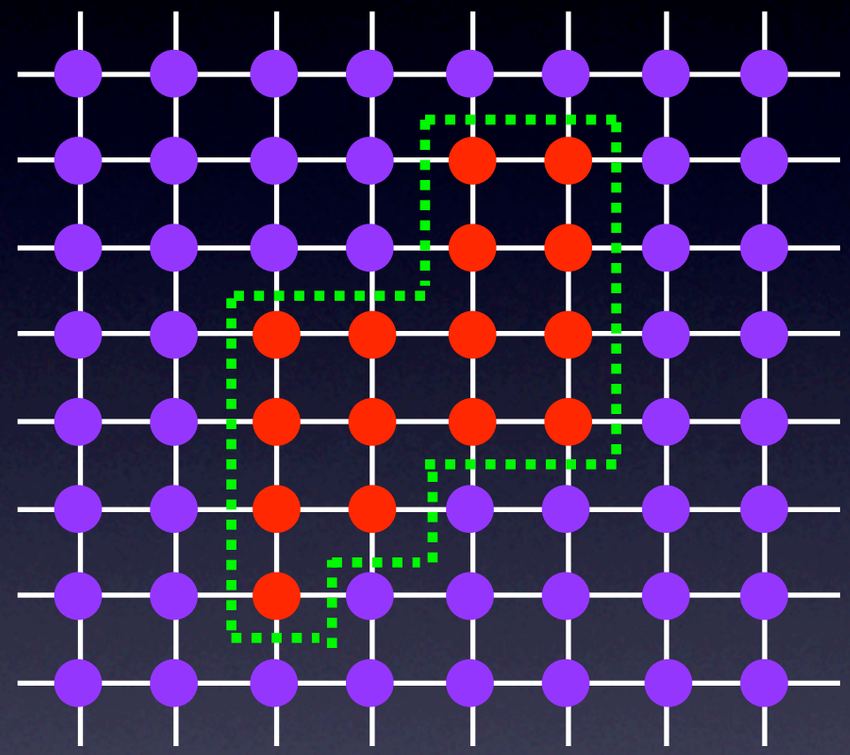
Calculate: free energy change to flip all spins



$E \propto \text{Constant}$

$\Delta F \approx 2J - T \log N$

entropy wins



$E \propto \text{Perimeter}$

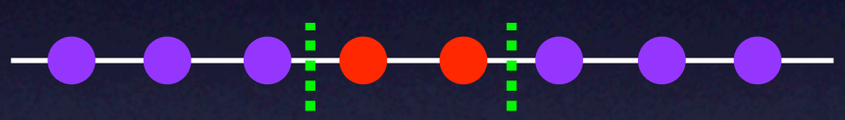
Self-Correcting

Calculate: free energy change to flip all spins

$\bullet = +1$

$\bullet = -1$

1D

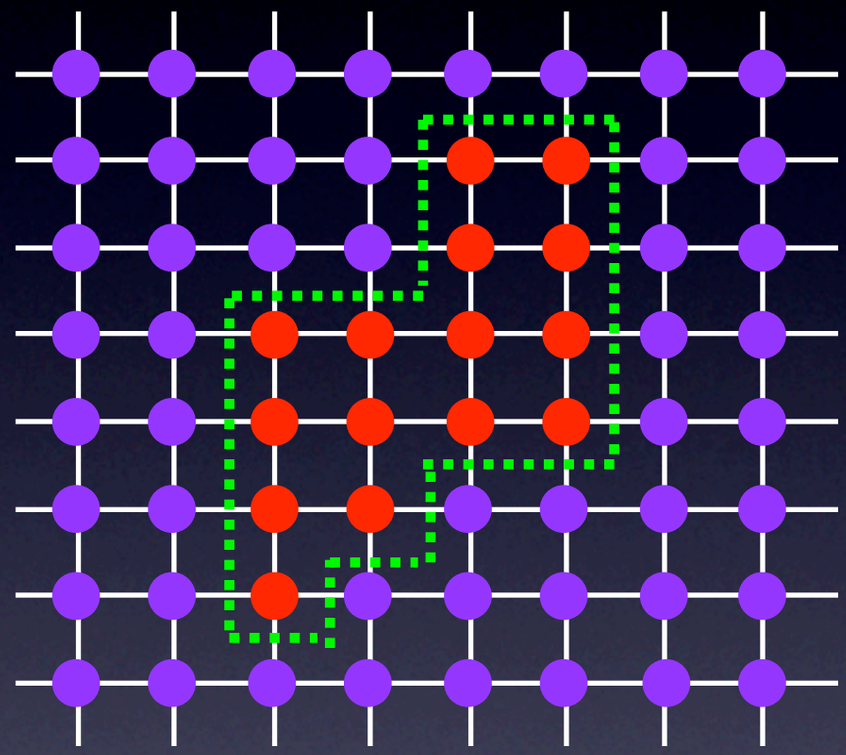


$E \propto \text{Constant}$

$\Delta F \approx 2J - T \log N$

entropy wins

2D



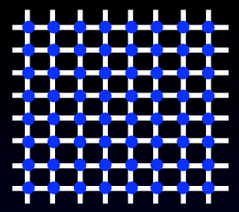
$E \propto \text{Perimeter}$

$\Delta F \approx 4NJ - T \log(N 2^{2N})$

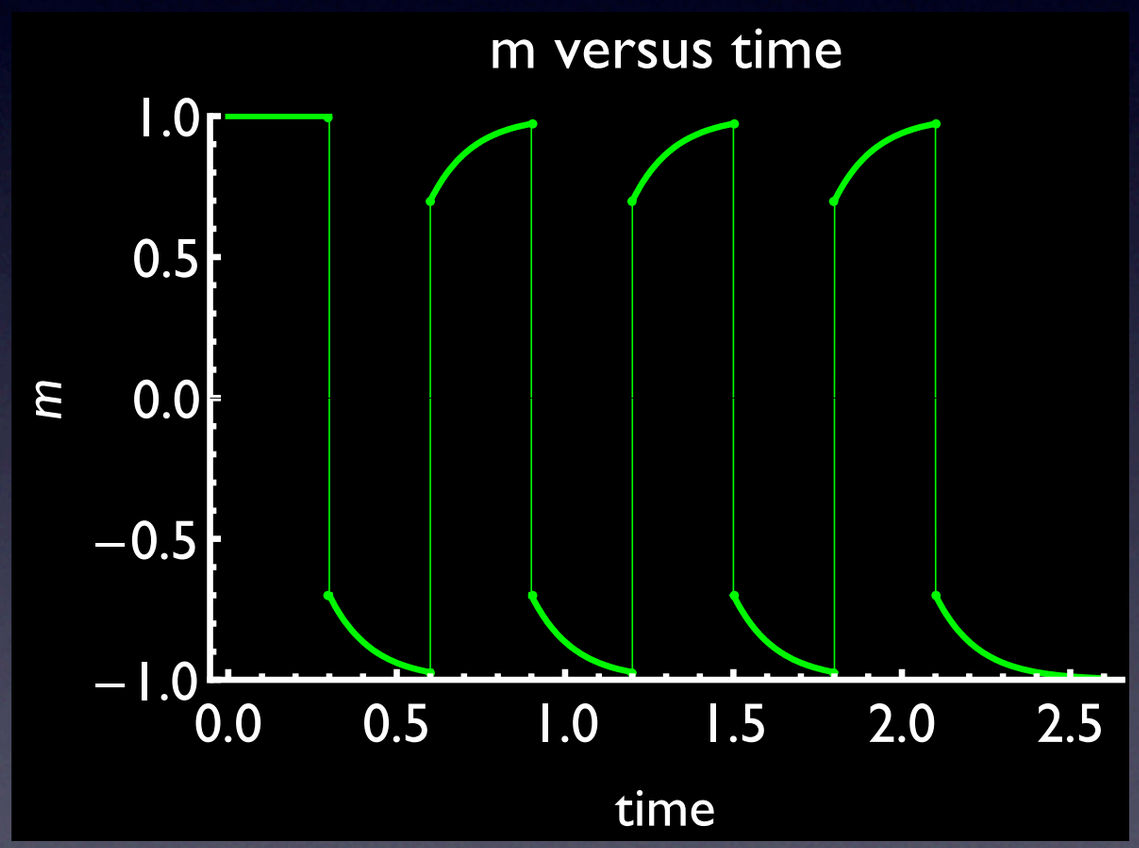
$T < T_C$ energy wins

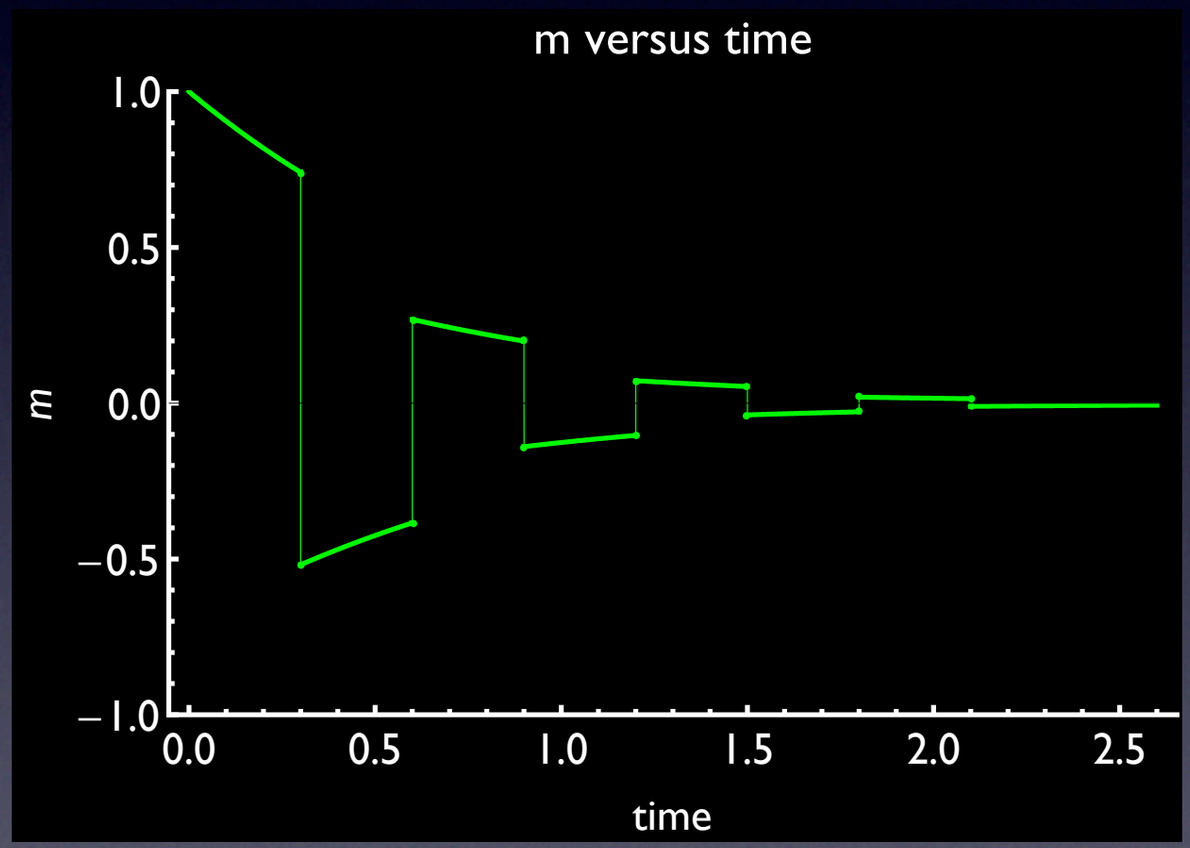
$T > T_C$ entropy wins

Fault-Tolerance

2D 

1D



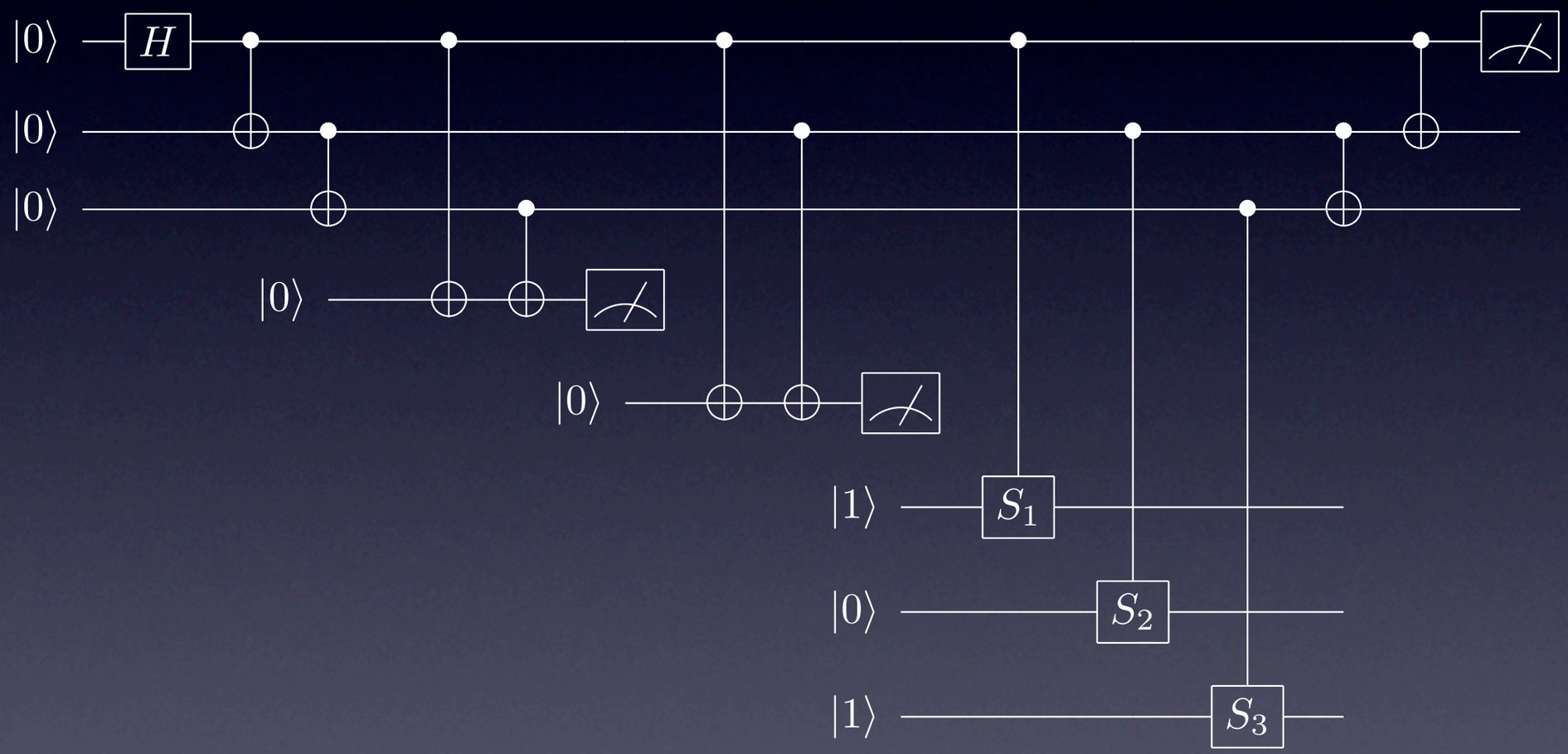




Dave? Weren't you going to be talking about quantum computing?

Quantum Computing

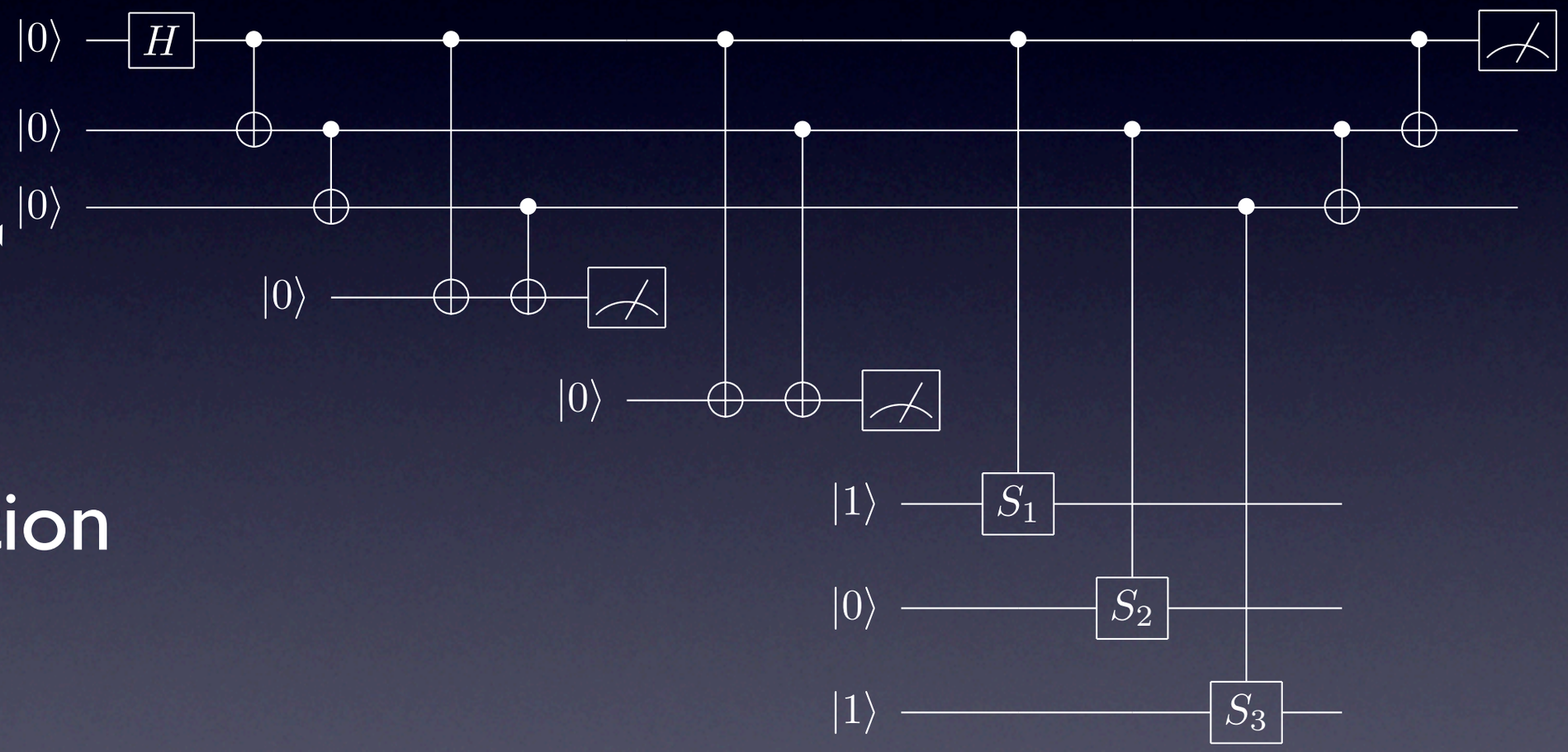
Circuit model for quantum algorithms



CS time ⟶

Quantum Computing

Circuit model for quantum algorithms

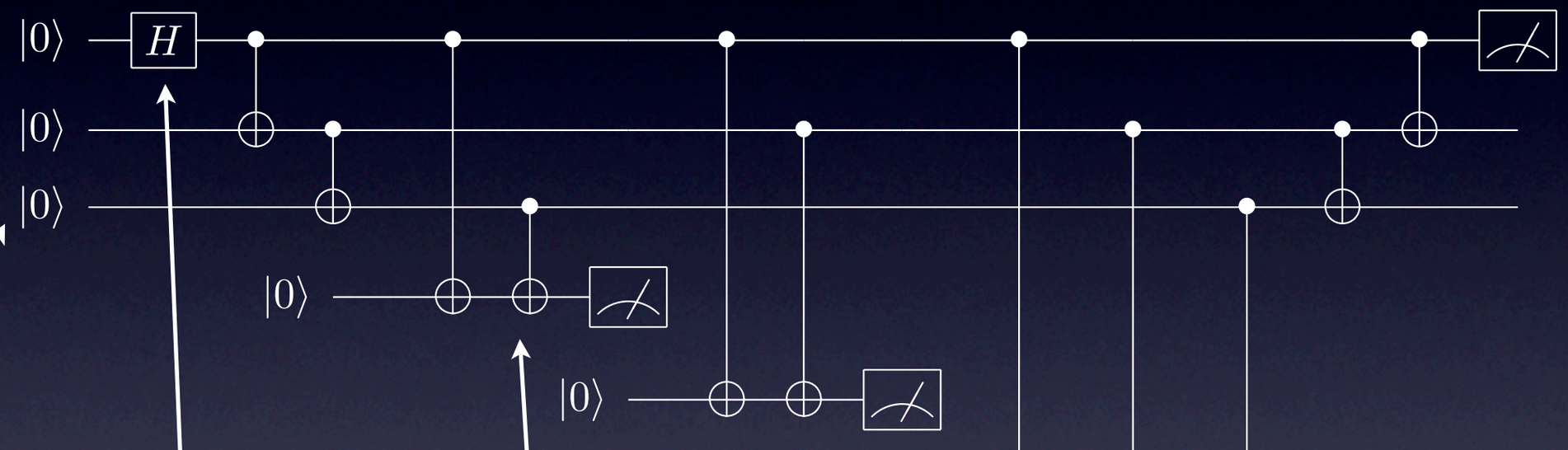


state preparation

CS time ⟶

Quantum Computing

Circuit model for quantum algorithms



state preparation

quantum gate

quantum gate

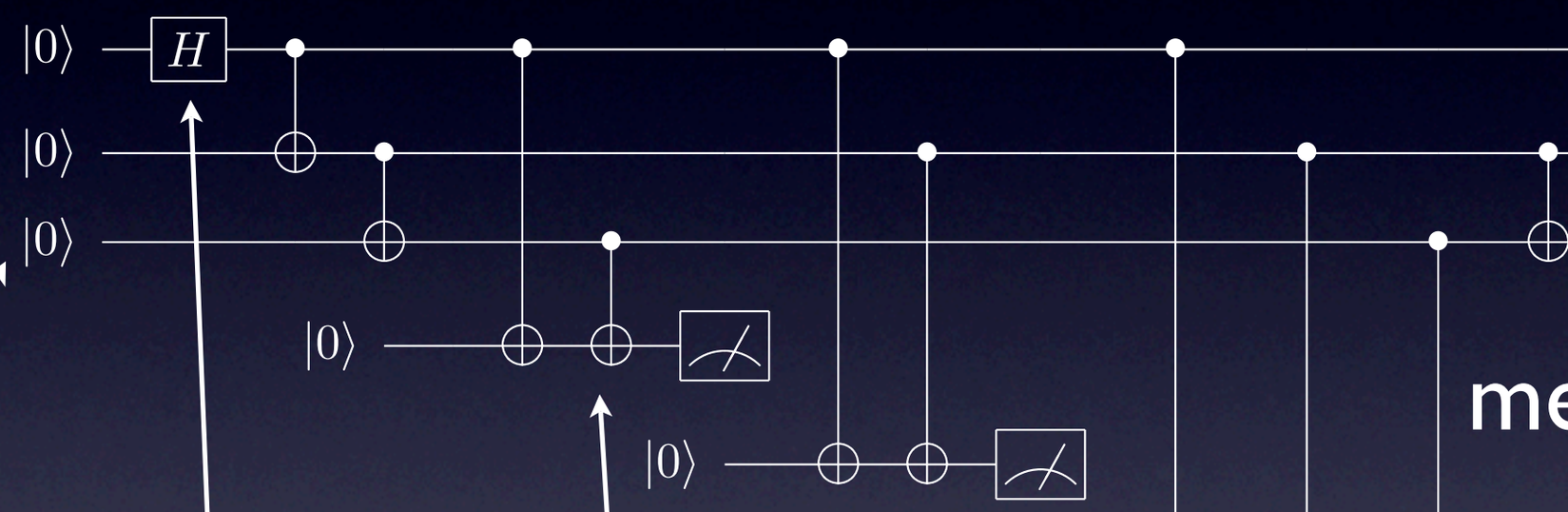
$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$C_X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

CS time $\longrightarrow$

Quantum Computing

Circuit model for quantum algorithms



state preparation

quantum gate

quantum gate

measurement

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$C_X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

CS time $\longrightarrow$

Quantum Algorithms

Factor

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

Quantum Algorithms

Factor

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

x

472772146107435302536223071973048224632914695302097116459852171130520711256363590397527

Quantum Algorithms

Factor

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

x

472772146107435302536223071973048224632914695302097116459852171130520711256363590397527

	Running time
Best classical algorithm	$\exp(N^{1/3})$
Shor's 1994 quantum algorithm	N^3

Breaks public key cryptosystems (RSA, Elliptic curves)

One of my Other Hats



Novel quantum algorithms

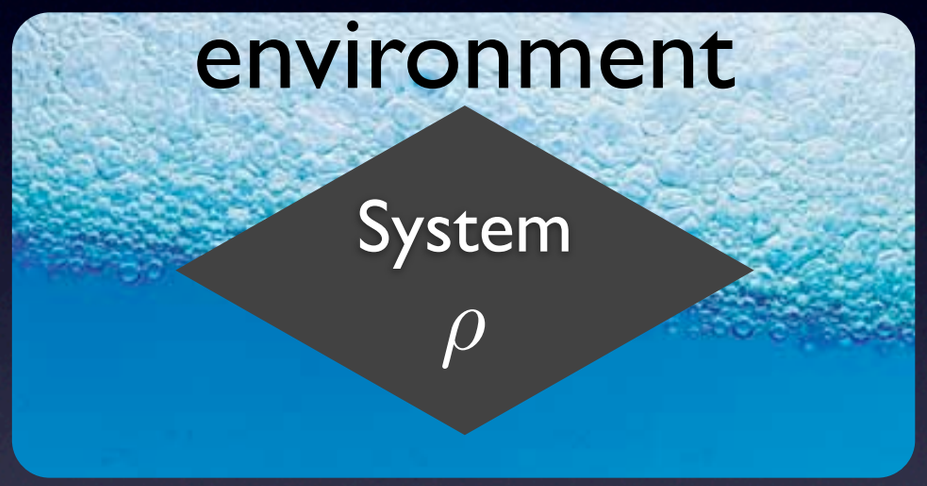
Obstacles

Decoherence:

$$(\alpha|0\rangle + \beta|1\rangle) \otimes |\psi_{\text{environment}}\rangle \rightarrow \alpha^*|0\rangle \otimes |\psi_0\rangle + \beta^*|1\rangle \otimes |\psi_1\rangle$$



pure



mixed

Infinite precision needed for gates?

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$



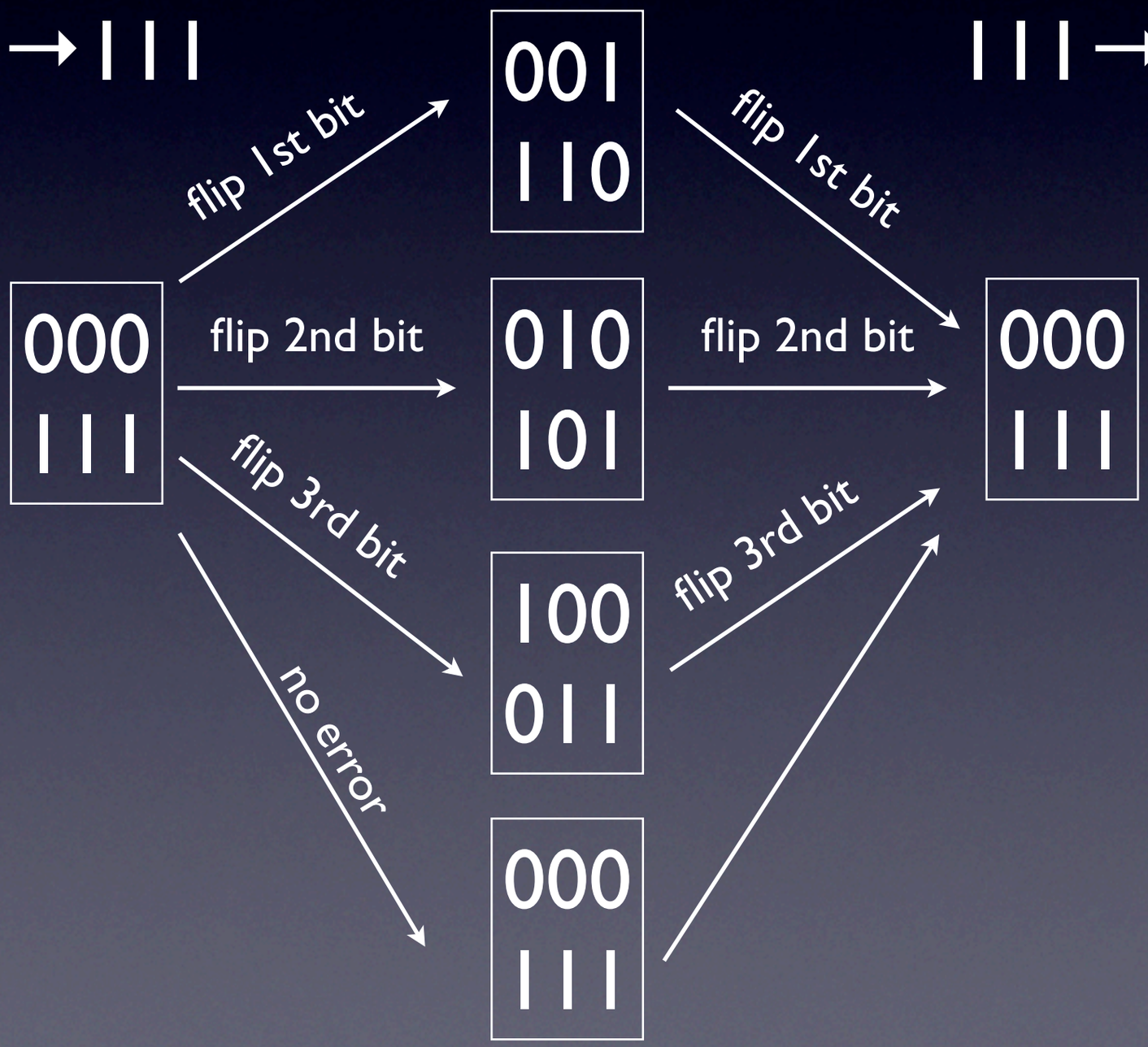
my
laser doesn't have a setting
for one over square root of
two

Error Correction

encode
0→000
1→111

diagnose

decode
000→0
111→1



Each bit flipped independently with probability p

$p_{fail}=3p^2-p^3$

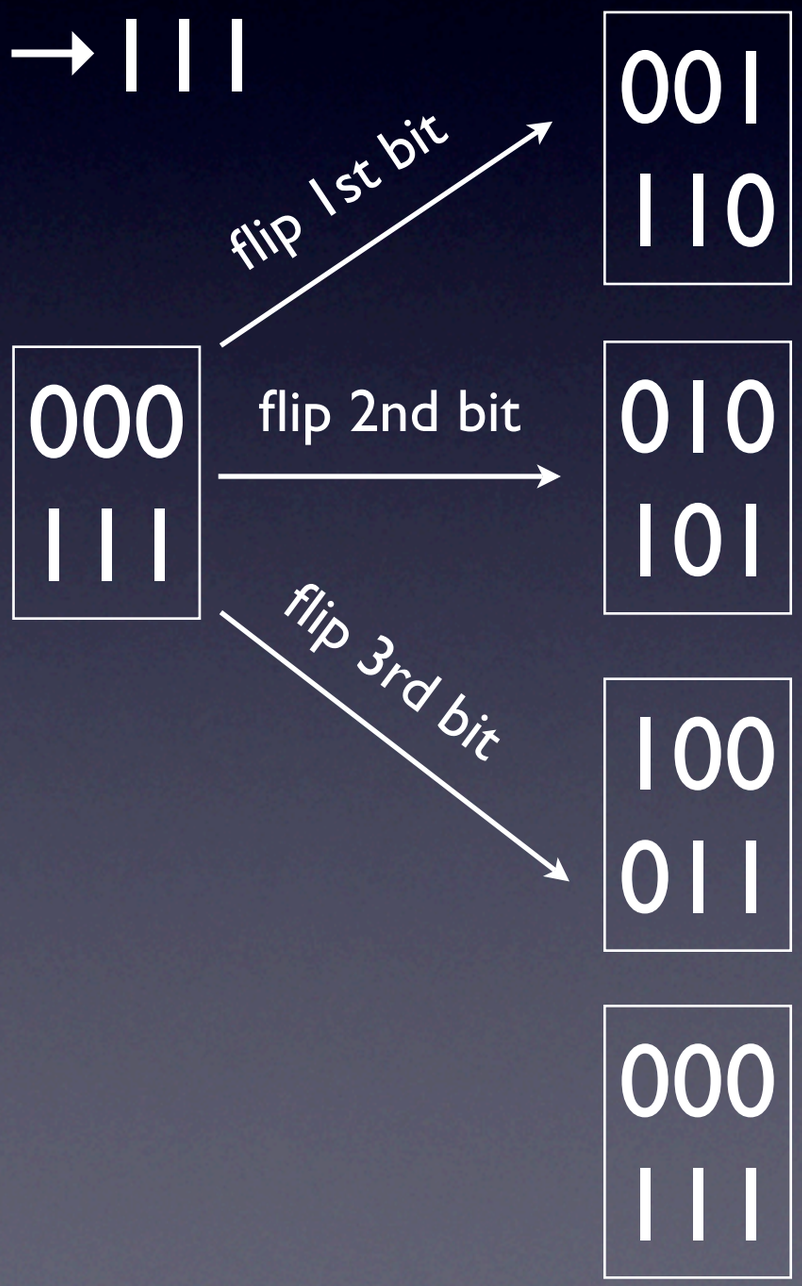
If $p<1/2$, then $p_{fail}<p$...success!

Blinded Error Correction

encode

0→000

1→111



	parity 1 and 2	parity 2 and 3
001 110	0	1
000 111		
010 101	1	1
100 011	1	0
000 111	0	0

diagnose which error occurred without revealing value of encoded information

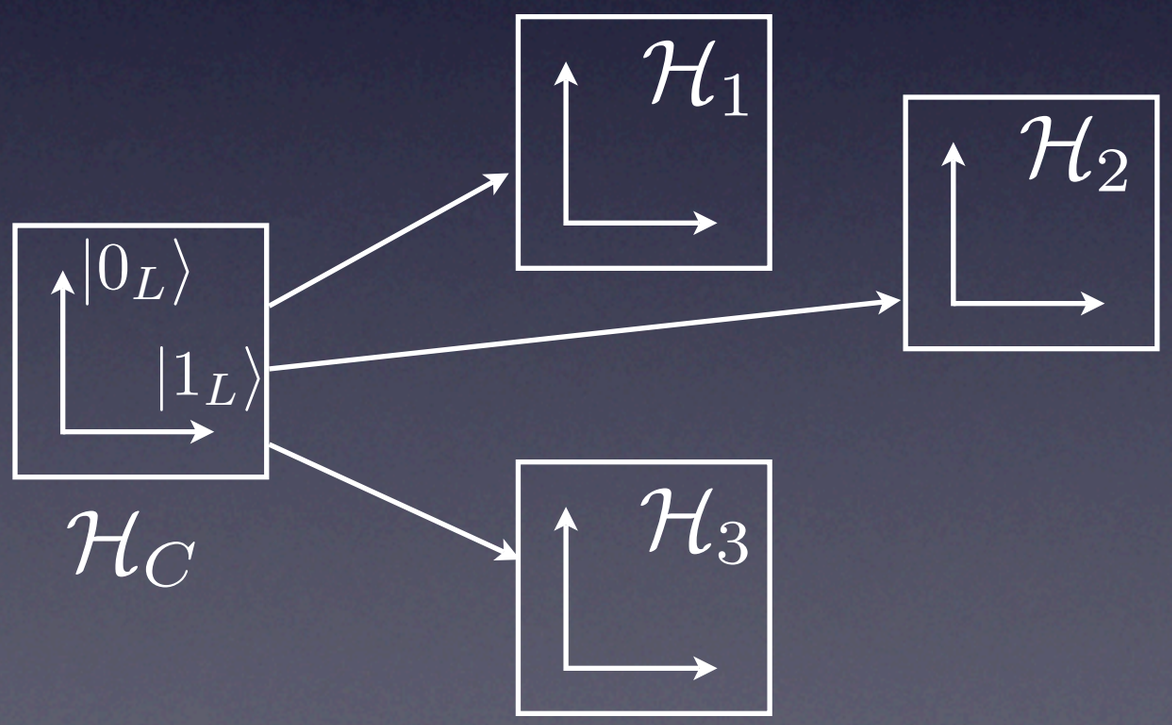
Quantum Error Correction

Basic idea: encoding information can help protect it



$$\mathcal{H} = [\mathbb{C}^2] \otimes [\mathbb{C}^2] \otimes [\mathbb{C}^2] \otimes [\mathbb{C}^2] \otimes [\mathbb{C}^2]$$

error correcting subspace $\mathcal{H}_C \subset \mathcal{H}$



errors take codespace to orthogonal subspaces

$$\mathcal{H}_C \perp \mathcal{H}_1 \perp \mathcal{H}_2 \perp \mathcal{H}_3$$

Quantum Error Correction



$$\mathcal{H} = [\mathbb{C}^2] \otimes [\mathbb{C}^2] \otimes [\mathbb{C}^2] \otimes [\mathbb{C}^2] \otimes [\mathbb{C}^2]$$

Commuting set of four operators

$$S_1 = X \otimes Z \otimes Z \otimes X \otimes I \qquad [S_i, S_j] = 0$$

$$S_2 = I \otimes X \otimes Z \otimes Z \otimes X \qquad S_i^2 = I$$

$$S_3 = X \otimes I \otimes X \otimes Z \otimes Z$$

$$S_4 = Z \otimes X \otimes I \otimes X \otimes Z$$

Basis: $|s_1, s_2, s_3, s_4, b\rangle \qquad s_i \in \{+1, -1\}, b \in \{+1, -1\}$

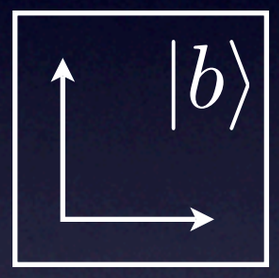
Codespace: $S_i|\psi_C\rangle = |\psi_C\rangle, |\psi_C\rangle \in \mathcal{H}_C$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Quantum Error Correction

$$S_1 = X \otimes Z \otimes Z \otimes X \otimes I$$
$$S_2 = I \otimes X \otimes Z \otimes Z \otimes X$$
$$S_3 = X \otimes I \otimes X \otimes Z \otimes Z$$
$$S_4 = Z \otimes X \otimes I \otimes X \otimes Z$$

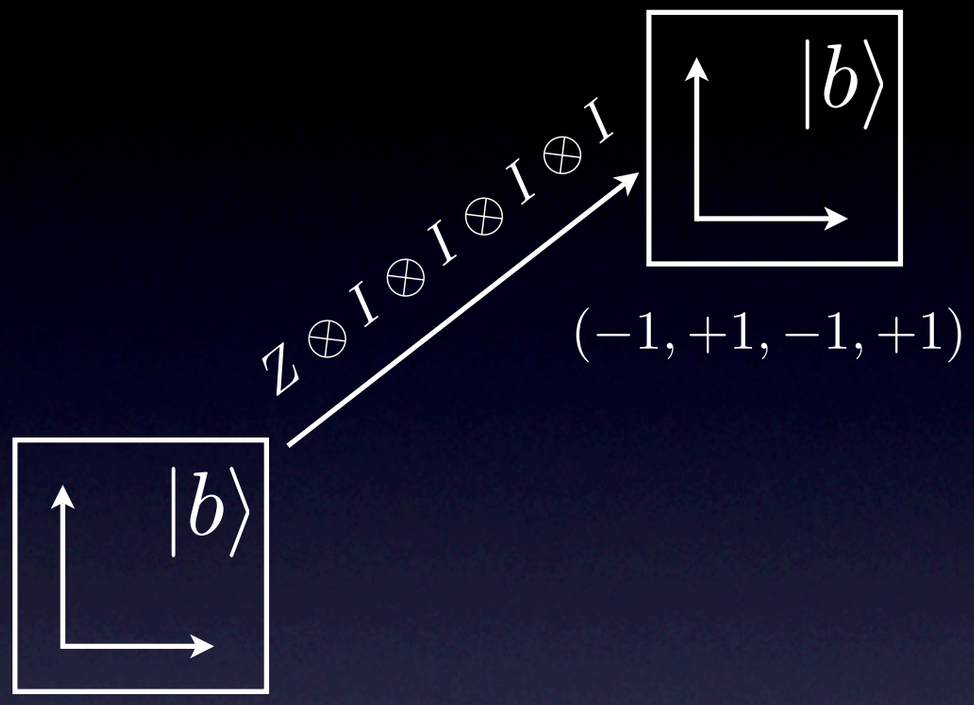
$$(s_1, s_2, s_3, s_4)$$



$\mathcal{H}_C$

$(+1, +1, +1, +1)$

Quantum Error Correction

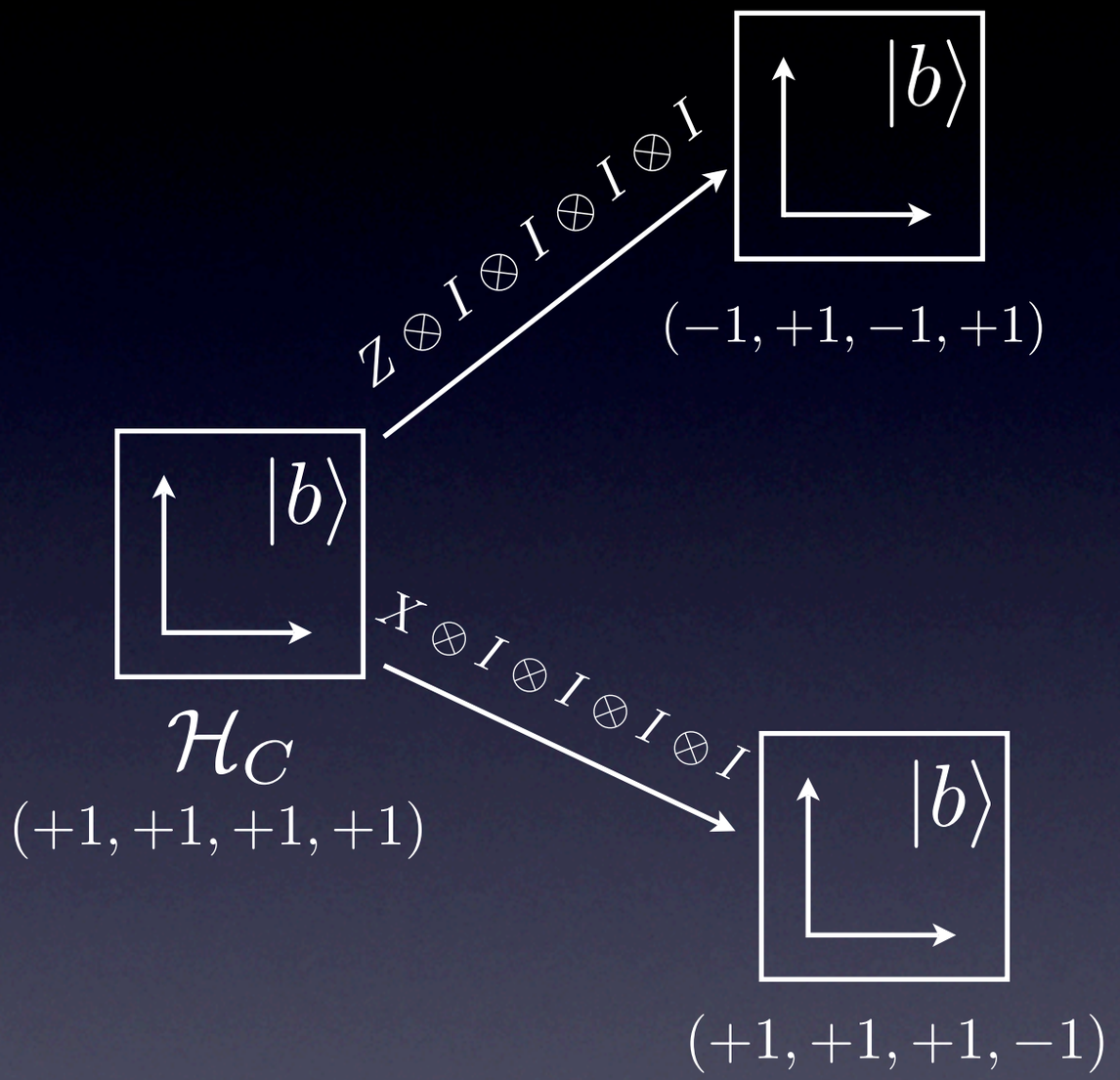


$$S_1 = X \otimes Z \otimes Z \otimes X \otimes I$$
$$S_2 = I \otimes X \otimes Z \otimes Z \otimes X$$
$$S_3 = X \otimes I \otimes X \otimes Z \otimes Z$$
$$S_4 = Z \otimes X \otimes I \otimes X \otimes Z$$

$$(s_1, s_2, s_3, s_4)$$

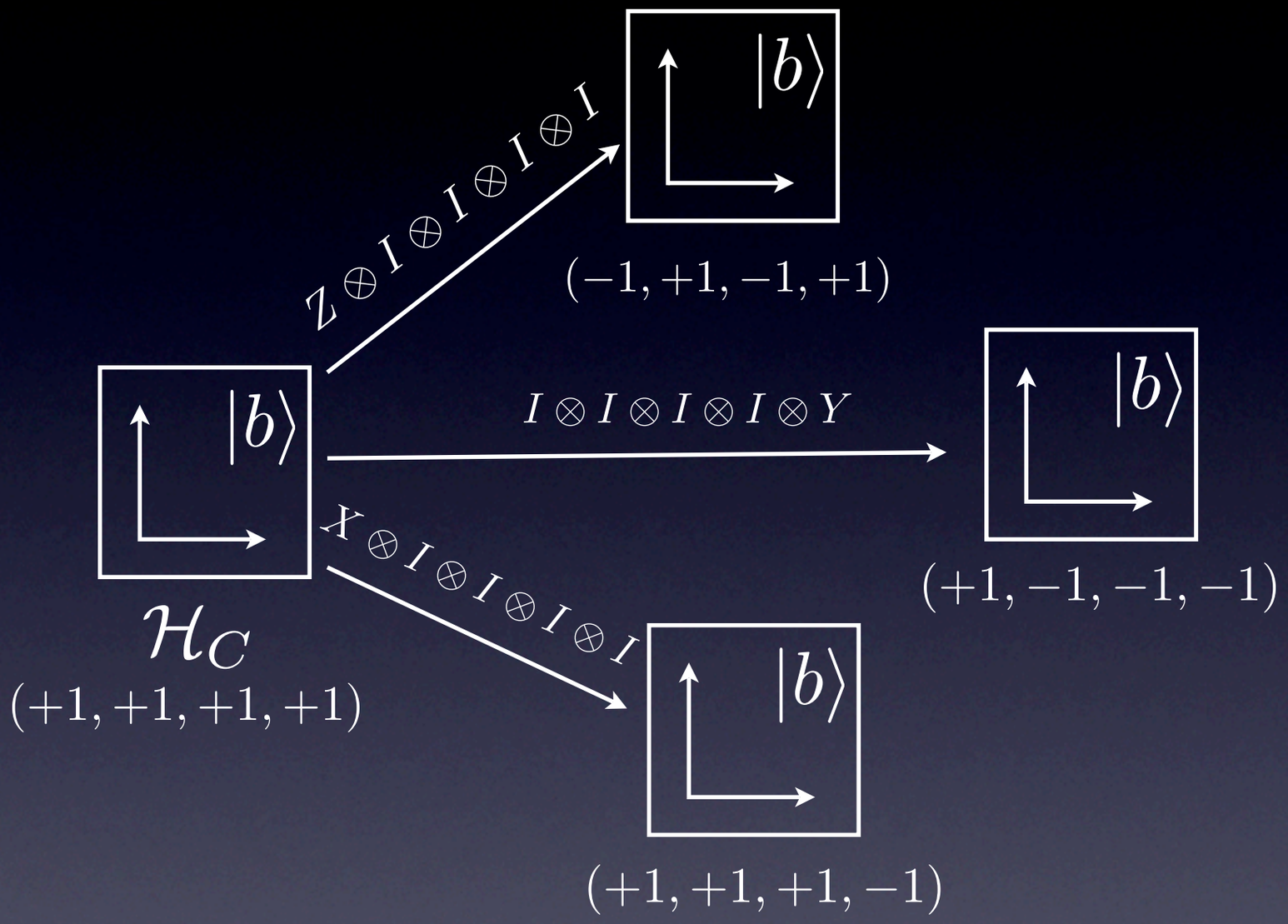
Quantum Error Correction

$$S_1 = X \otimes Z \otimes Z \otimes X \otimes I$$
$$S_2 = I \otimes X \otimes Z \otimes Z \otimes X$$
$$S_3 = X \otimes I \otimes X \otimes Z \otimes Z$$
$$S_4 = Z \otimes X \otimes I \otimes X \otimes Z$$
$$(s_1, s_2, s_3, s_4)$$



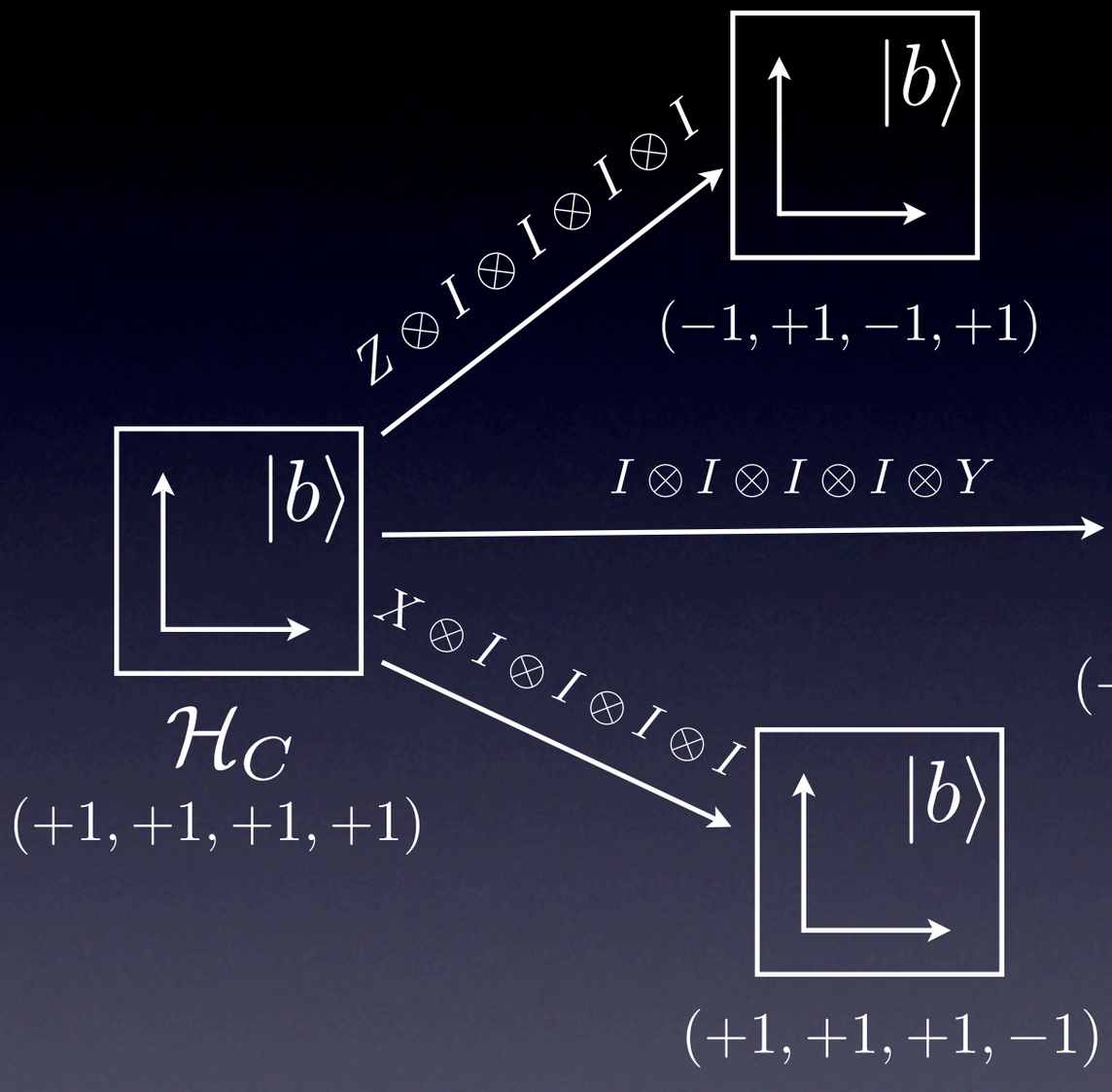
Quantum Error Correction

$$S_1 = X \otimes Z \otimes Z \otimes X \otimes I$$
$$S_2 = I \otimes X \otimes Z \otimes Z \otimes X$$
$$S_3 = X \otimes I \otimes X \otimes Z \otimes Z$$
$$S_4 = Z \otimes X \otimes I \otimes X \otimes Z$$
$$(s_1, s_2, s_3, s_4)$$



Quantum Error Correction

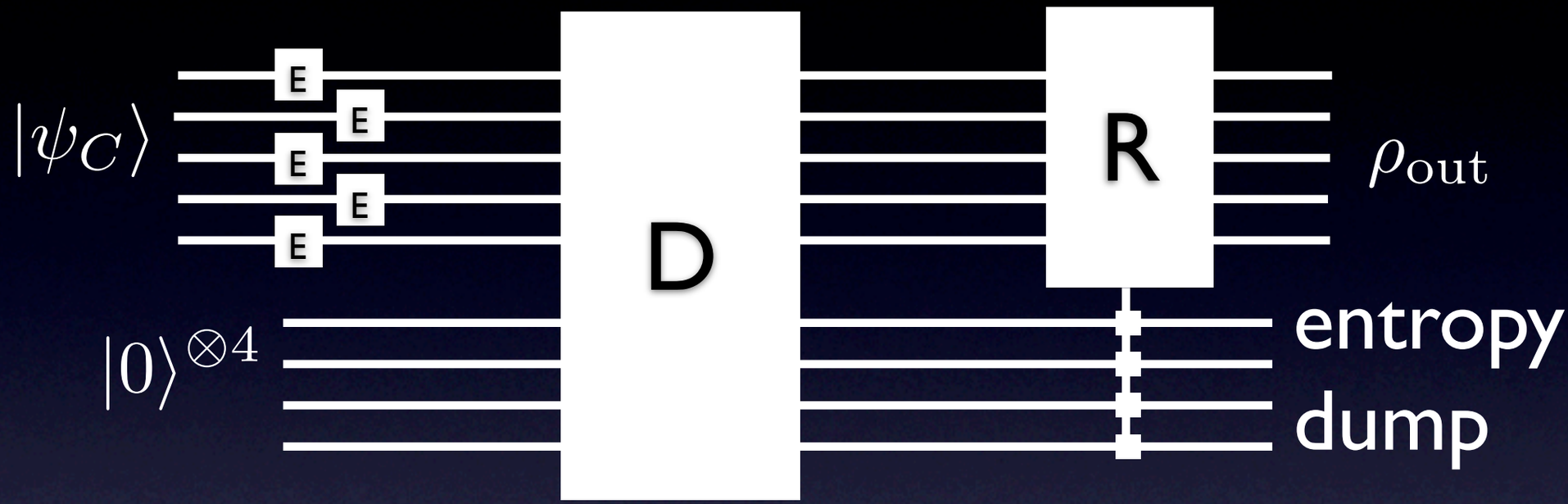
$$S_1 = X \otimes Z \otimes Z \otimes X \otimes I$$
$$S_2 = I \otimes X \otimes Z \otimes Z \otimes X$$
$$S_3 = X \otimes I \otimes X \otimes Z \otimes Z$$
$$S_4 = Z \otimes X \otimes I \otimes X \otimes Z$$
$$(s_1, s_2, s_3, s_4)$$



Every operator which acts nontrivially on only one qubit moves to orthogonal subspace

We can diagnose single qubit errors without destroying quantum information by measuring S_i

Quantum Restorative Organ



If local transition error transition amplitudes are ϵ then output states has error amplitude $C\epsilon^2$

Note: a continuous set of error are corrected by the above procedure (quantum theory is linear!)

$$\sum_{i=0}^{15} \alpha_i E_i$$

Fault-Tolerant Quantum Computing

Consider quantum computing where all quantum circuit components can fail:



state preparation



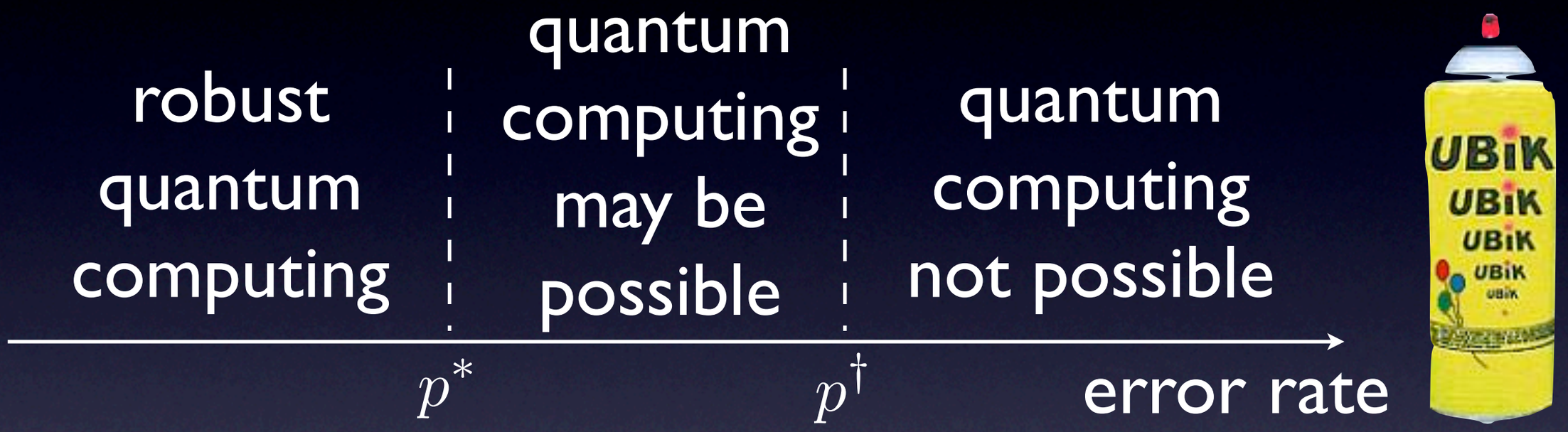
gates

wires



measurements

The Quantum Threshold Theorem



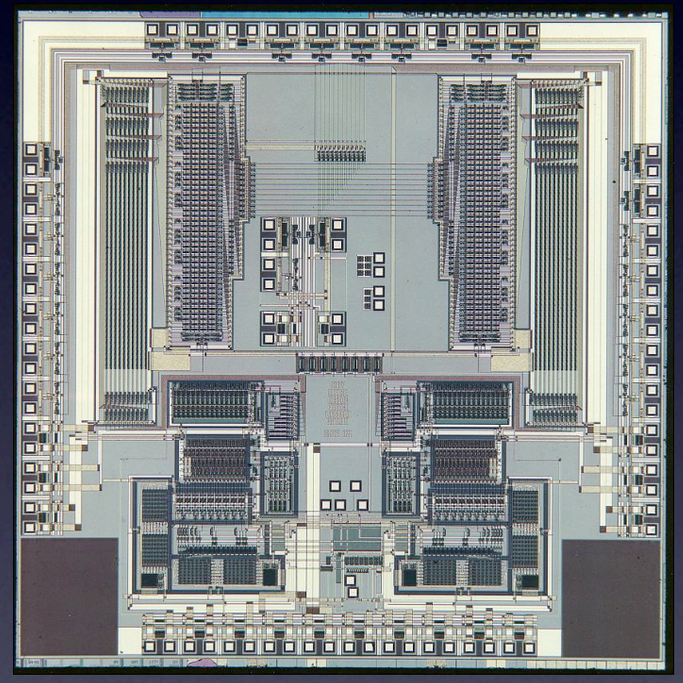
Theorem: If decoherence and lack of control are below a threshold p* (relative to coherent control), then a quantum computation of size g can be carried out with error probability ε using O(g log(g/ε)) extra qubits

Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Shor, Kitaev, Gottesman, Preskill (~1996)

The Race

But....





Kitaev's Idea



Alexei Kitaev

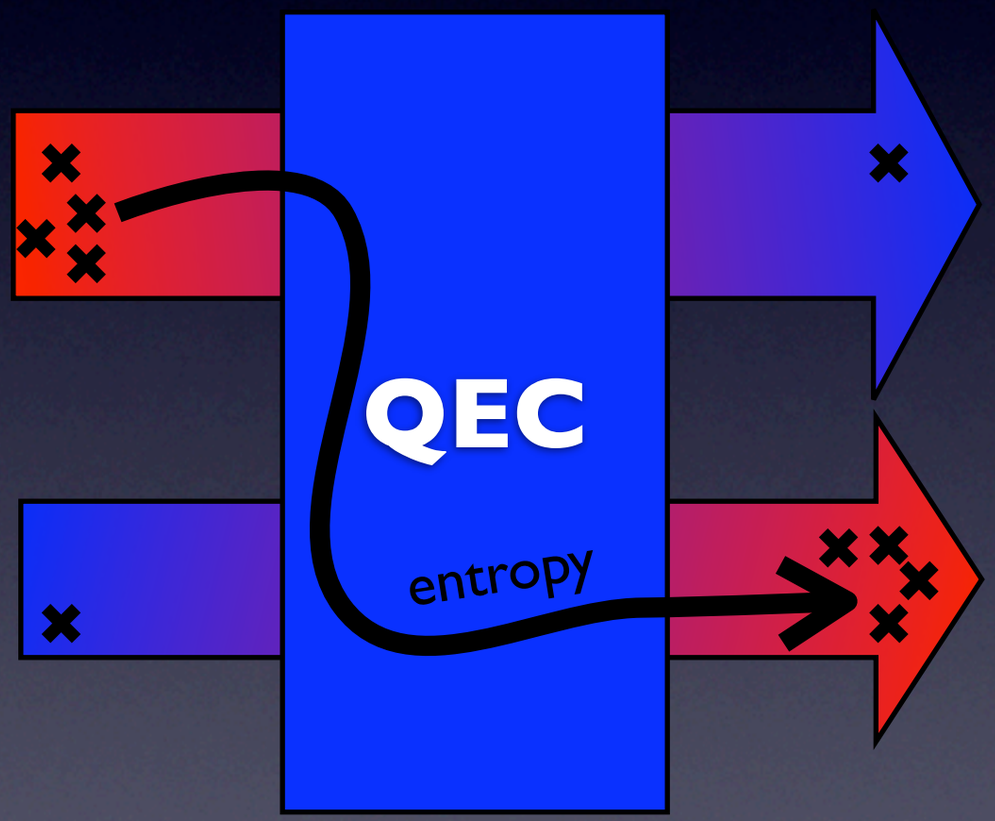
"Magnetism arise from spins of individual atoms. Each spin is quite sensitive to thermal fluctuations. But the spins interact with each other and tend to be oriented in the same direction. If some spin flips to the opposite direction, the interaction forces it to flip back to the direction of other spins. This process is quite similar to the standard error correction procedure for the repetition code. We may say that errors are being corrected at the physical level. Can we propose something similar in the quantum case? **Yes, but it is not so simple.**"

Kitaev, "Fault-tolerant quantum computation by anyons" (1997, published 2003)

Quantum Error Correction is Refrigeration

noisy quantum data

cold ancillas

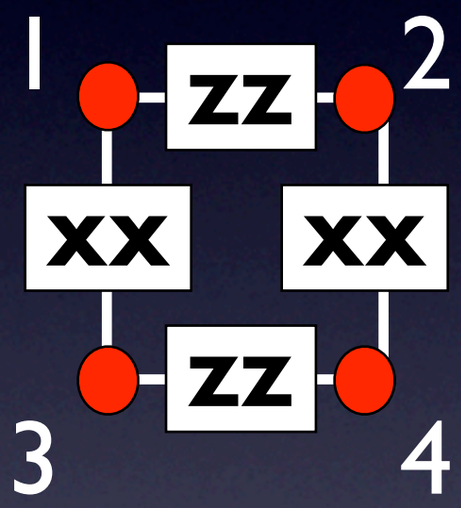


clean quantum data

hot ancillas

Energetic Protection

Bacon, Ph.D. thesis, "Decoherence, Control, and Symmetry" (2001)



$$H = -\Delta(X_1X_3 + X_2X_4 + Z_1Z_2 + Z_3Z_4)$$

[[4,2,2]] QEC Code

stabilizer

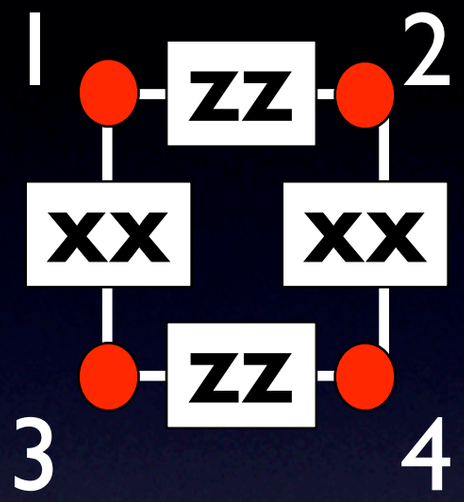
$$S_1 = X_1X_2X_3X_4$$
$$S_2 = Z_1Z_2Z_3Z_4$$

logical qubits

$$\bar{X}_1 = X_1X_3 \quad \bar{X}_2 = X_1X_2$$
$$\bar{Z}_1 = Z_1Z_3 \quad \bar{Z}_2 = Z_1Z_2$$

basis: $|S_1, S_2, z_1, z_2\rangle$

Energetic Protection

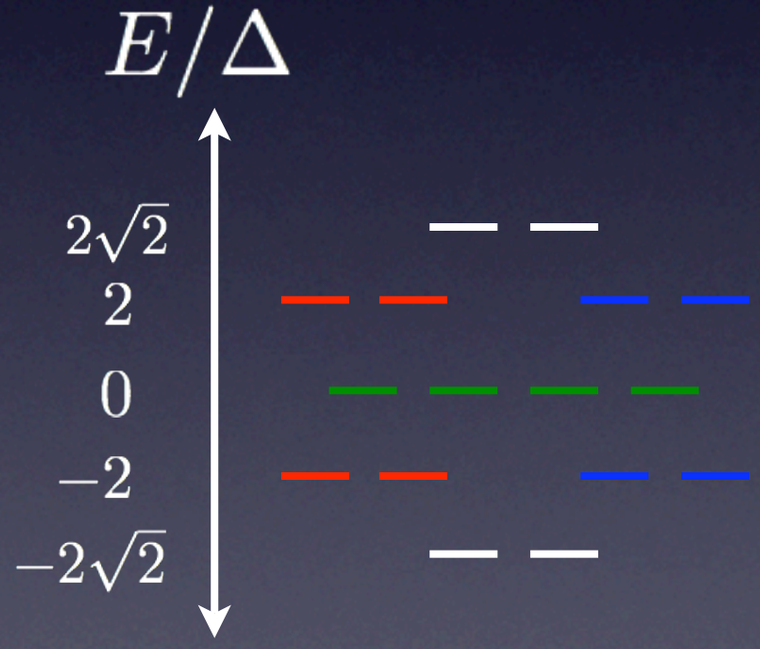


$$H = -\Delta(X_1X_3 + X_2X_4 + Z_1Z_2 + Z_3Z_4)$$

$$H = -\Delta(\bar{X}_1(I + S_1) + \bar{Z}_1(I + S_2))$$

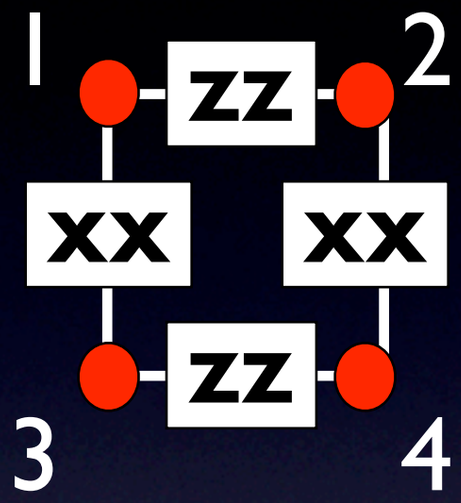
$|S_1, S_2, z_1, z_2\rangle$

S_1	S_2	H	
$+1$	$+1$	$-2\Delta(\bar{X}_1 + \bar{Z}_1)$	
$+1$	-1	$-2\Delta\bar{Z}_1$	
-1	$+1$	$-2\Delta\bar{X}_1$	
-1	-1	0	



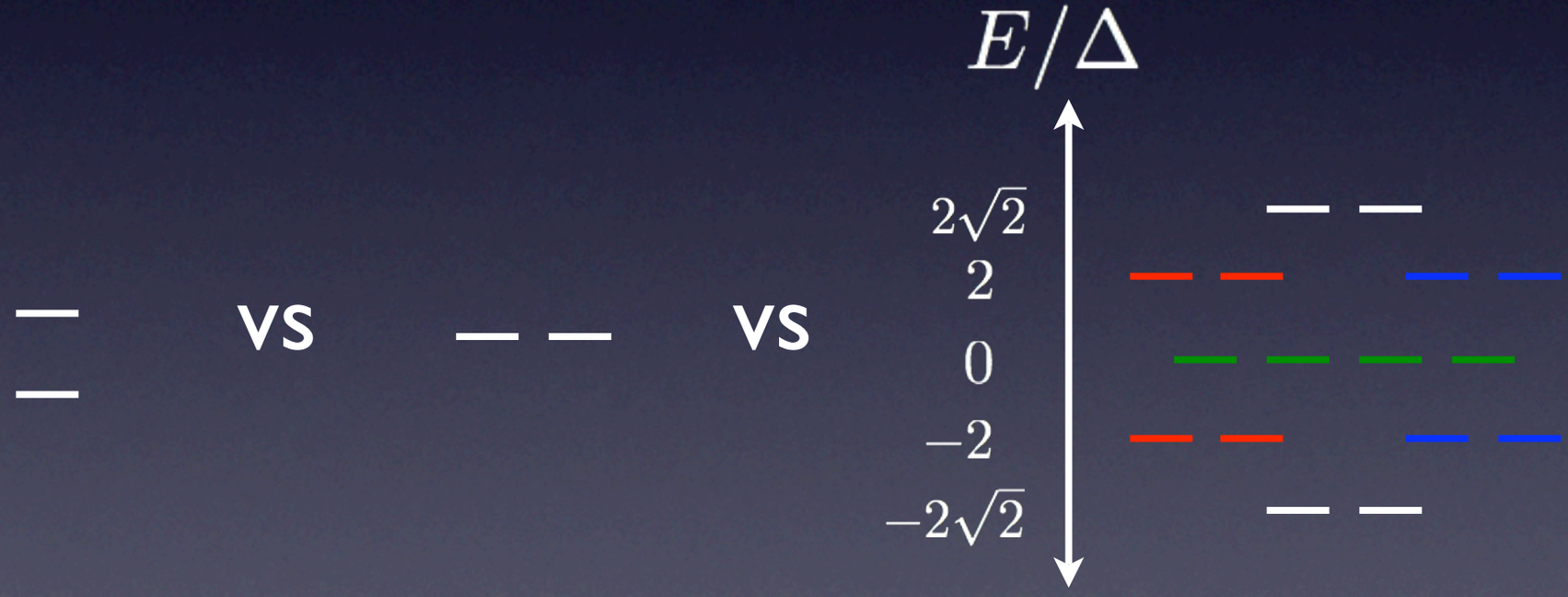
Ground state is QEC Code

Supercoherence



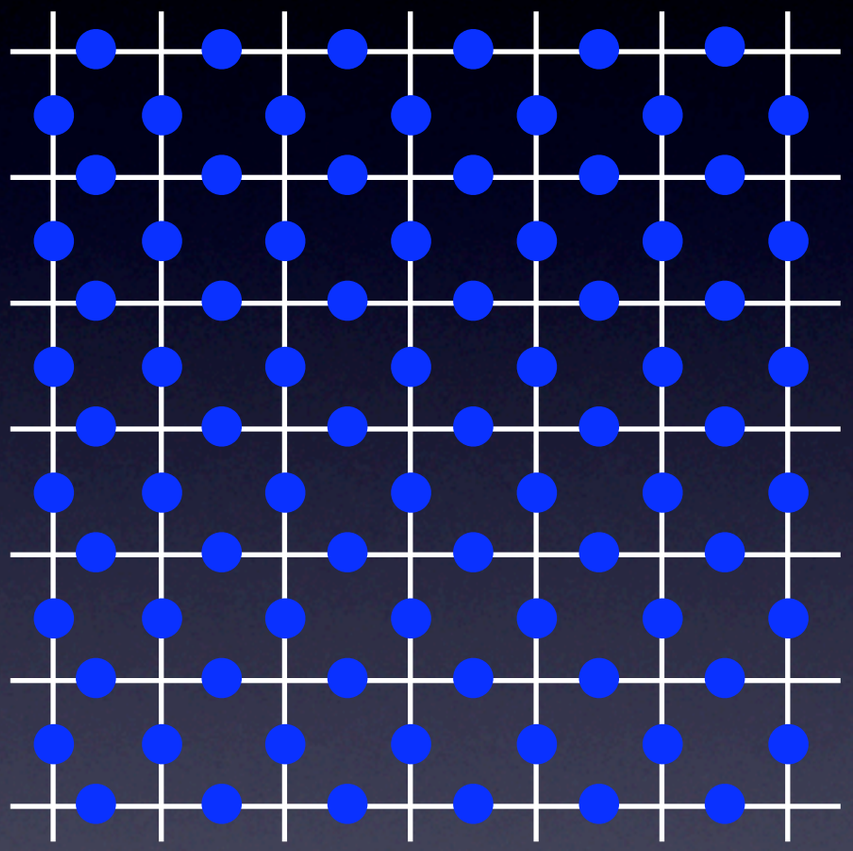
$$H = -\Delta(X_1X_3 + X_2X_4 + Z_1Z_2 + Z_3Z_4)$$

All single qubit errors excite out of ground state



Bacon, Brown, and Whaley, "Coherence-Preserving Quatum Bits" (2001)

Kitaev's Toric System



periodic boundary conditions (torus)

plaquette operator



$$S_p = \prod_{e \in \delta P} Z_e$$

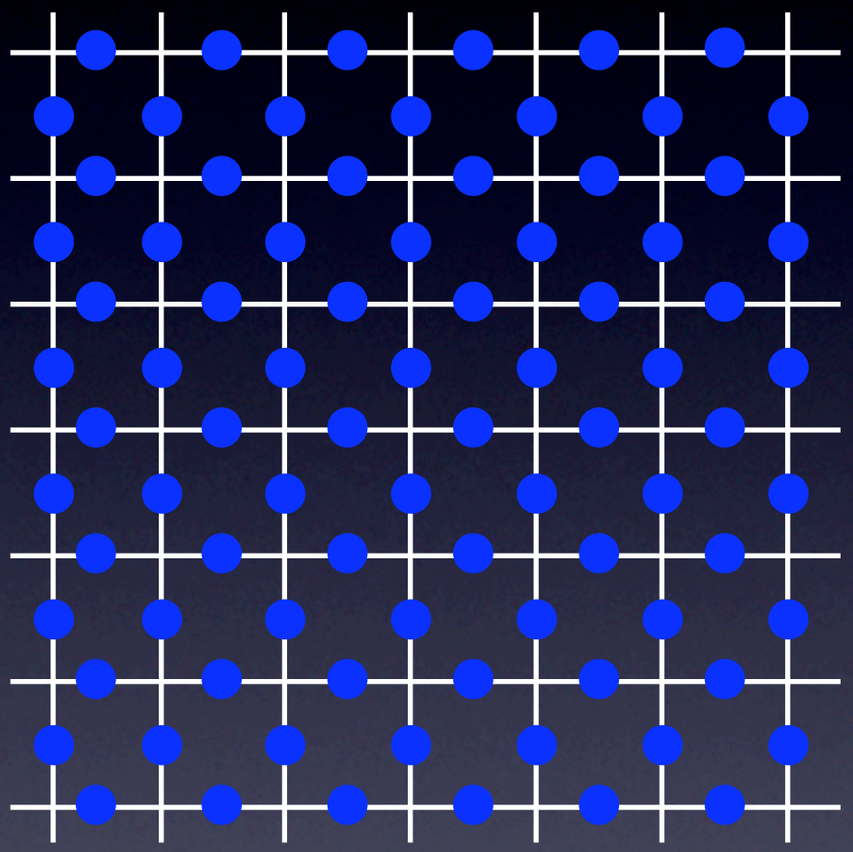
vertex operator



$$S_v = \prod_{e|v \in e} X_e$$

$$[S_p, S_v] = 0 \qquad S_p^2 = S_v^2 = I$$

Kitaev's Toric System



periodic boundary conditions (torus)

plaquette operator



$$S_p = \prod_{e \in \delta P} Z_e$$

vertex operator



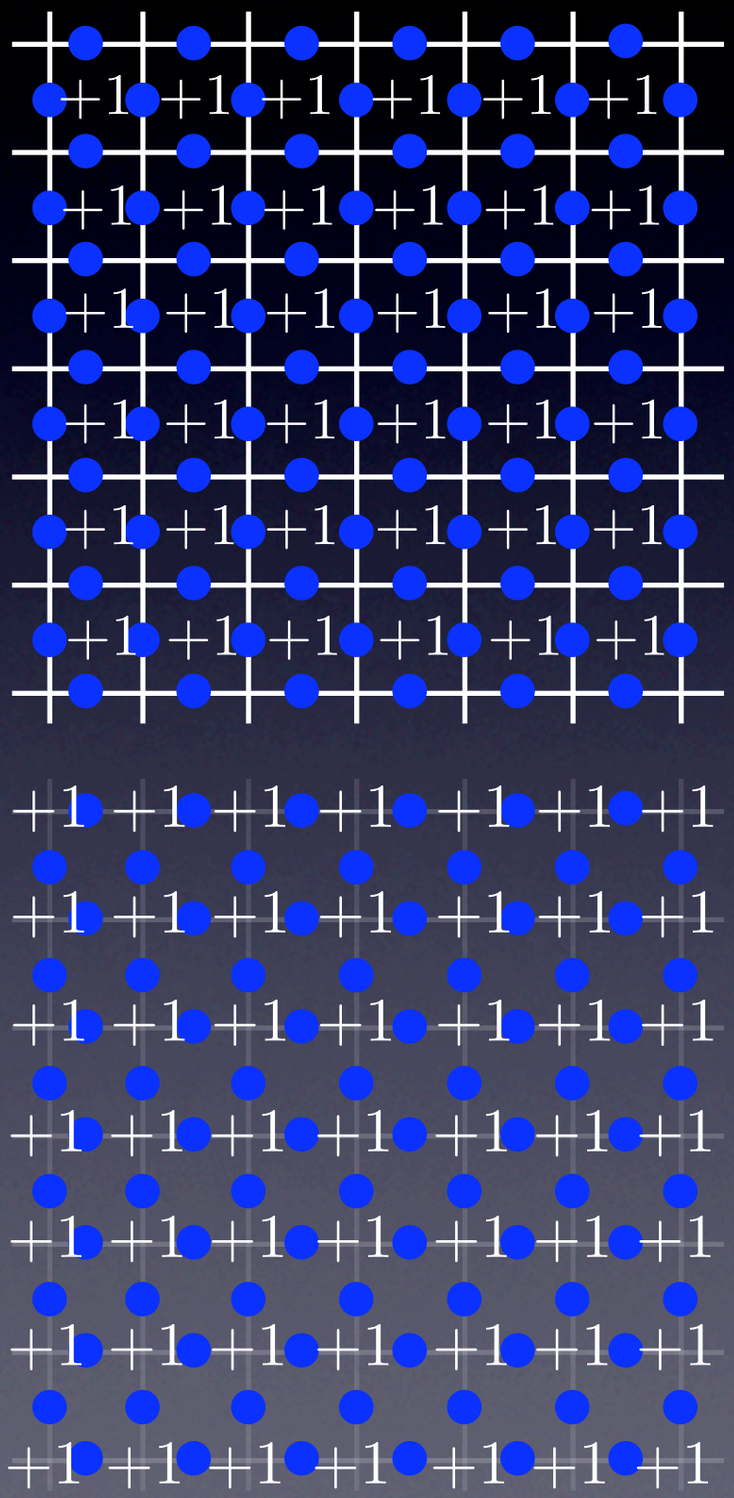
$$S_v = \prod_{e|v \in e} X_e$$

$$[S_p, S_v] = 0 \qquad S_p^2 = S_v^2 = I$$

Kitaev Hamiltonian:

$$H = -\frac{\Delta}{4}\left[\sum_p S_p + \sum_v S_v\right]$$

Kitaev's Toric System



$$H = -\frac{\Delta}{4}\left[\sum_p S_p + \sum_v S_v\right]$$

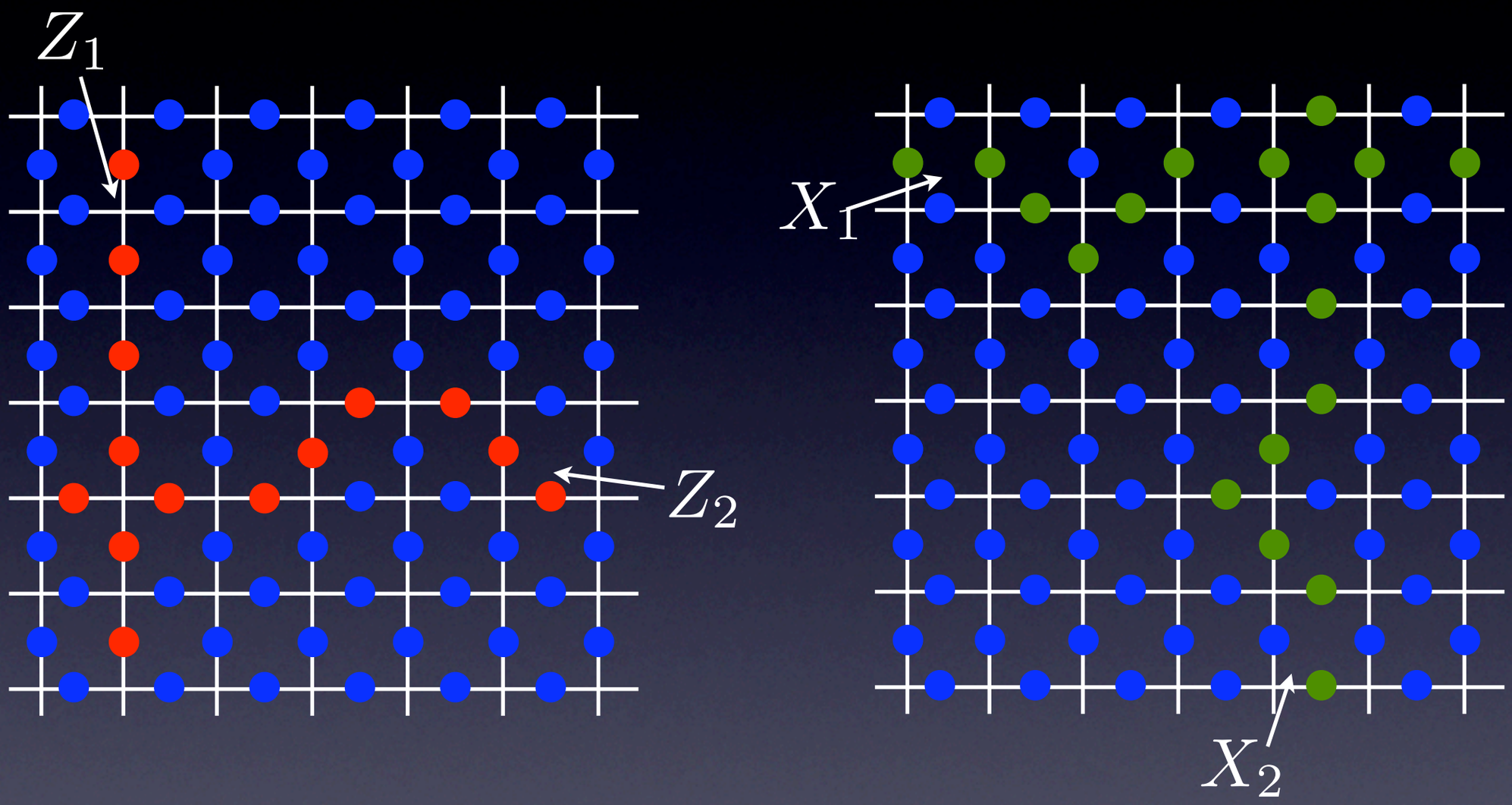
Ground state has

$$S_p|\psi\rangle = |\psi\rangle$$

$$S_v|\psi\rangle = |\psi\rangle$$

Degeneracy?

Kitaev's Toric System



operators which loop around an odd number of times commute with S_p, S_v but not all among each other

ground state is 4-fold degenerate

Topological Quantum Computing



non-abelian Anyons can be used to build a universal quantum computer

Kitaev, "Fault-tolerant quantum computation by anyons" (1997, published 2003)
Freedman, "Quantum computation and the localization of modular functors" (2001)

Topological Quantum Computing



non-abelian Anyons can be used to build a universal quantum computer

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Kitaev, "Fault-tolerant quantum computation by anyons" (1997, published 2003)
Freedman, "Quantum computation and the localization of modular functors" (2001)

Topological Quantum Computing



non-abelian Anyons can be used to build a universal quantum computer

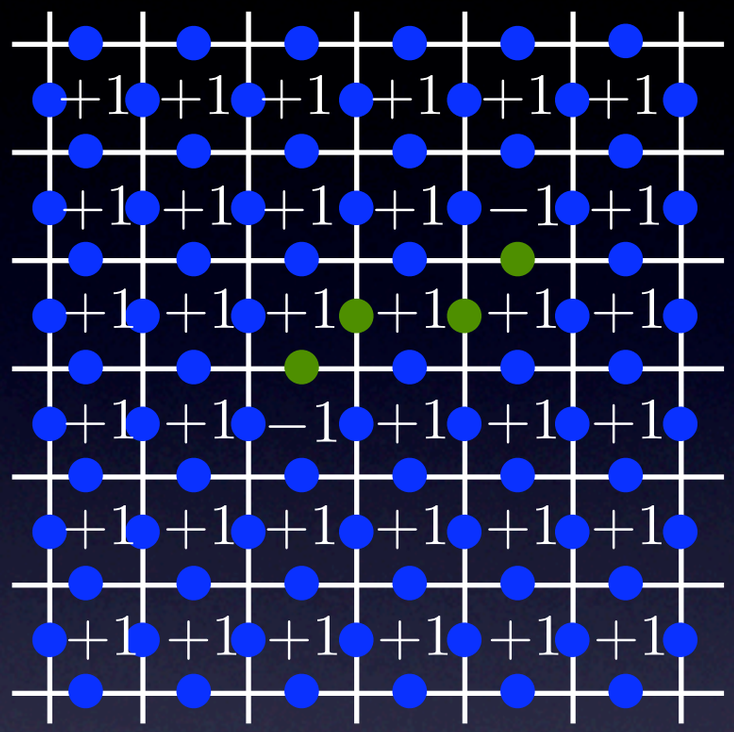
$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$



quantum computing using representations of braid group

Kitaev, "Fault-tolerant quantum computation by anyons" (1997, published 2003)
Freedman, "Quantum computation and the localization of modular functors" (2001)

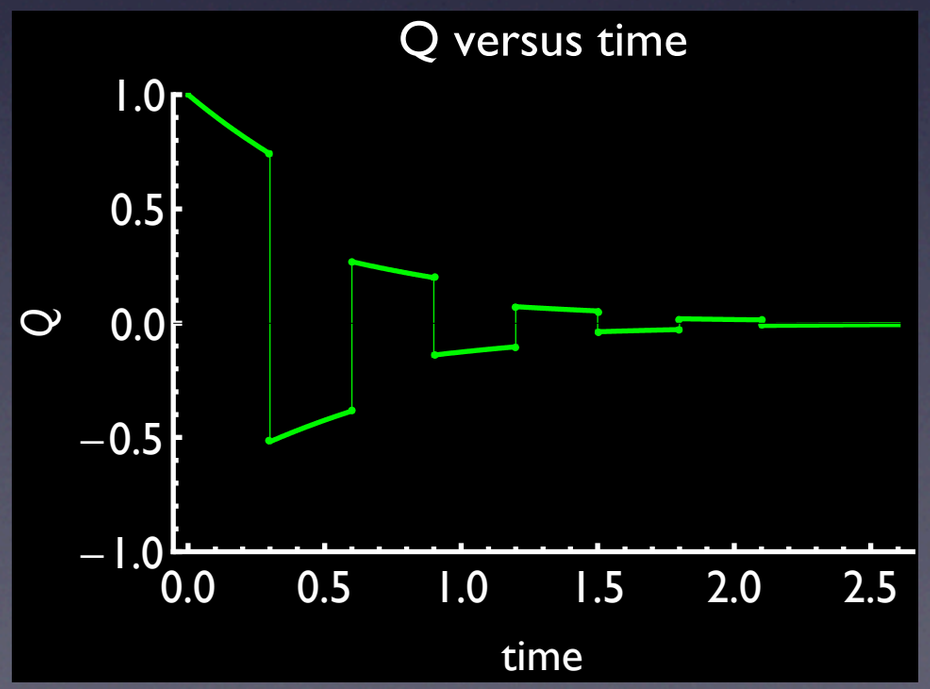
Kitaev's Toric Protection



End of chain of X operators on dual lattice only changes S_p at the end of the chain

Excitations are particles

Like 1D Ising model:
good for memory but not self correcting



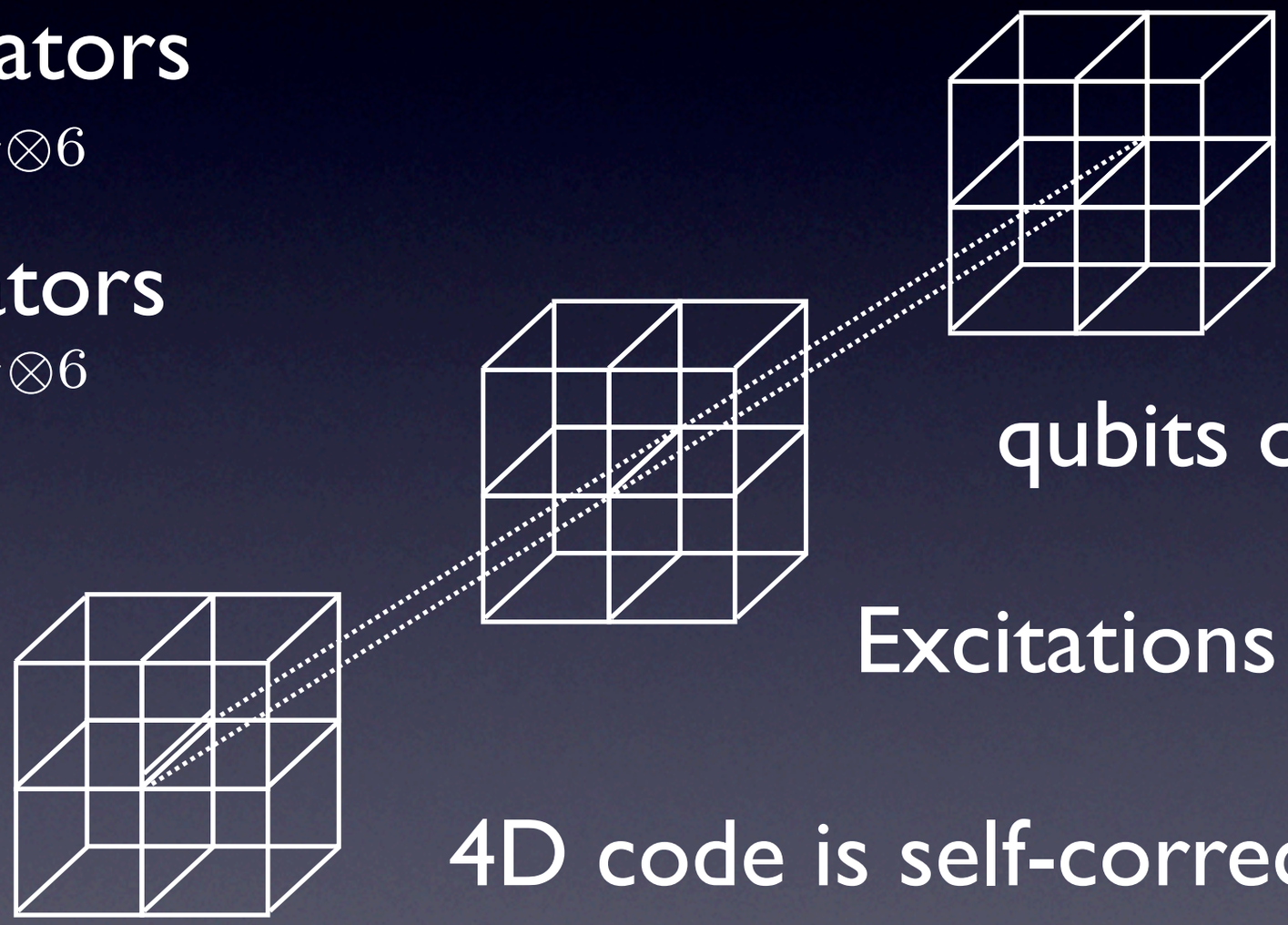
Self Correcting Quantum Memory

cube operators

$$S_c = Z^{\otimes 6}$$

link operators

$$S_l = X^{\otimes 6}$$



qubits on plaquettes

Excitations are strings

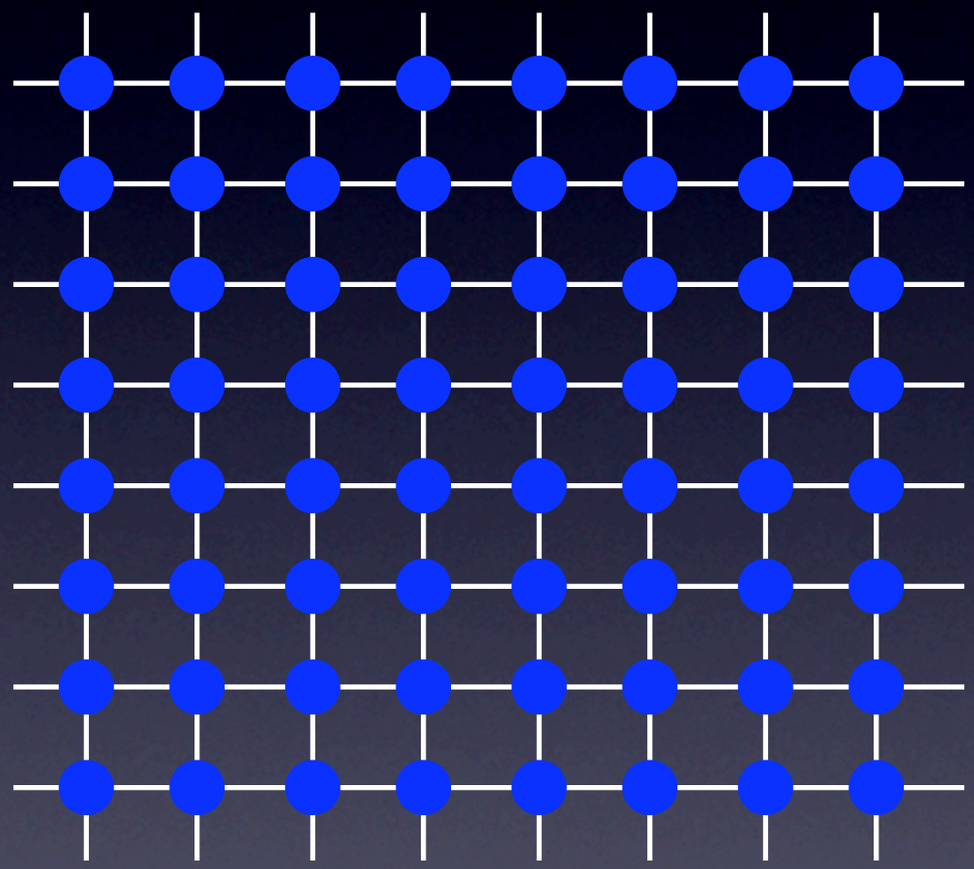
4D code is self-correcting

Dennis, Kitaev, Landahl, Preskill "Topological quantum memory" (2002)

Major Open Question

- Do there exists self-correcting quantum memories in realistic settings?

Quantum Compass Model



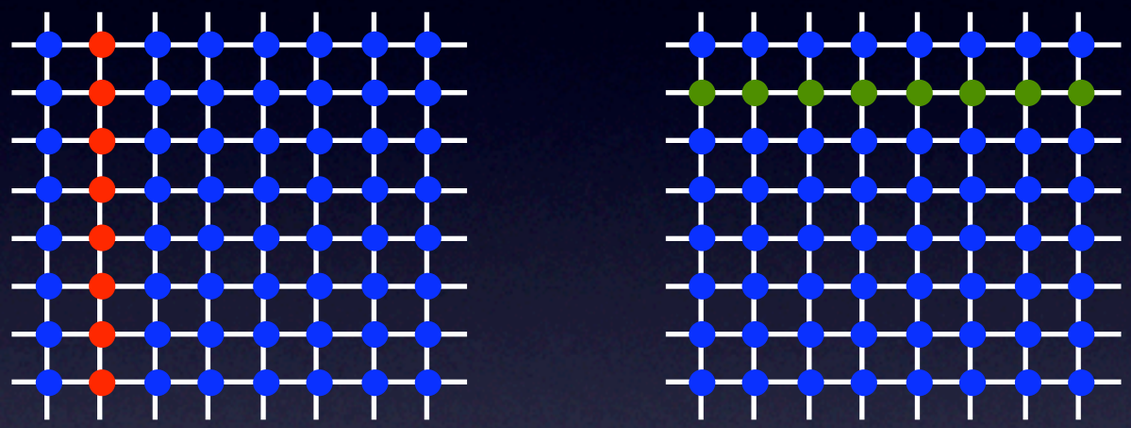
$$H = \sum_{i,j} X_{i,j} X_{i+1,j} + Z_{i,j} Z_{i,j+1}$$



Dorier, Becca, Mila "The Quantum Compass Model on a Square Lattice" (2005)

Quantum Compass Model

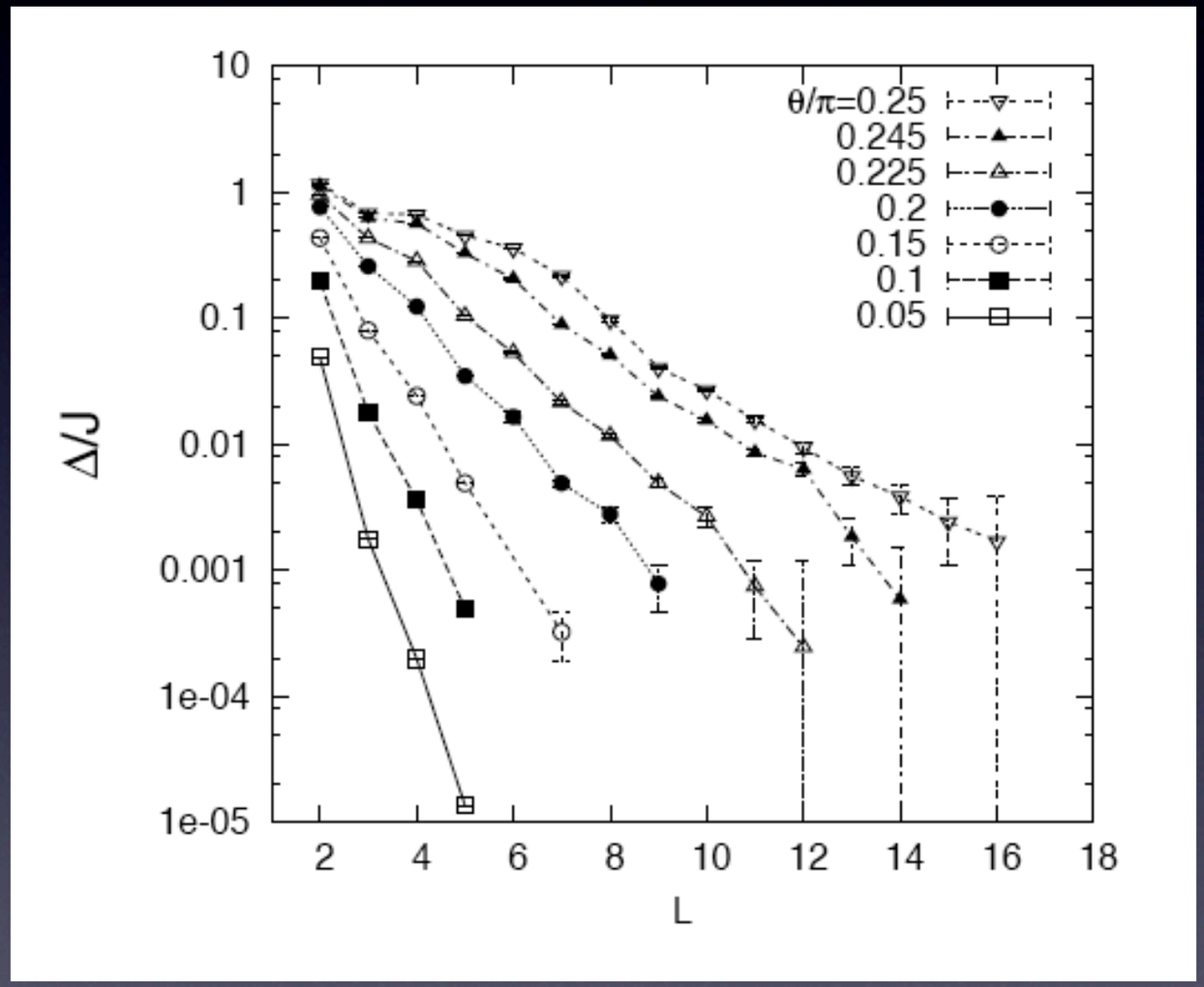
Operators which commute with Hamiltonian:



imply ground state is two-fold degenerate

Can it be used to protect quantum information?

Quantum Compass Model



$\tan(\theta)$

= ratio of XX and ZZ interactions

Length of square lattice

Dorier, Becca, Mila "The Quantum Compass Model on a Square Lattice" (2005)

Subsystem Codes

Take a subspace $\mathcal{H}_S \subset \mathcal{H}$ and put a tensor product on it

$$\mathcal{H}_S = \mathcal{H}_L \otimes \mathcal{H}_G$$

encode quantum information into $\mathcal{H}_L$

One doesn't care what happens to information in $\mathcal{H}_G$

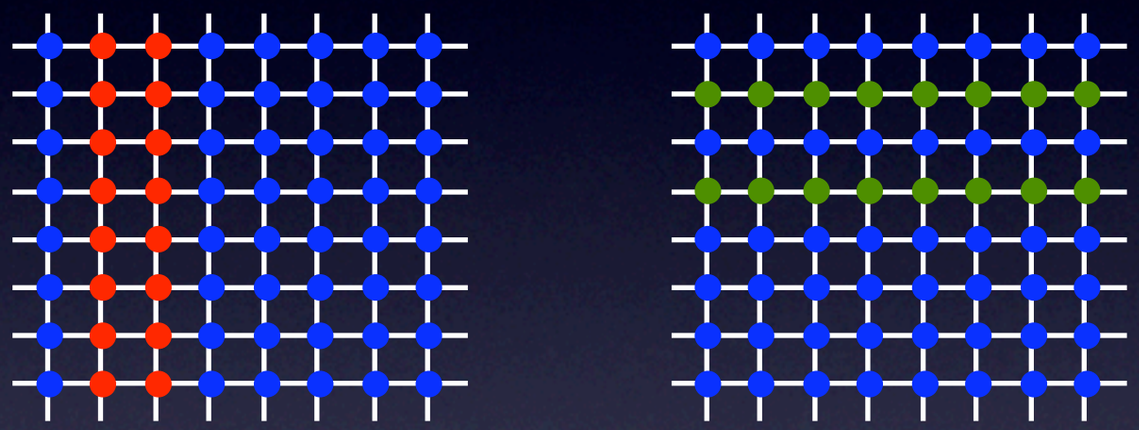
Quantum information protected up to change in "gauge"

Compass Model Subsystem Code

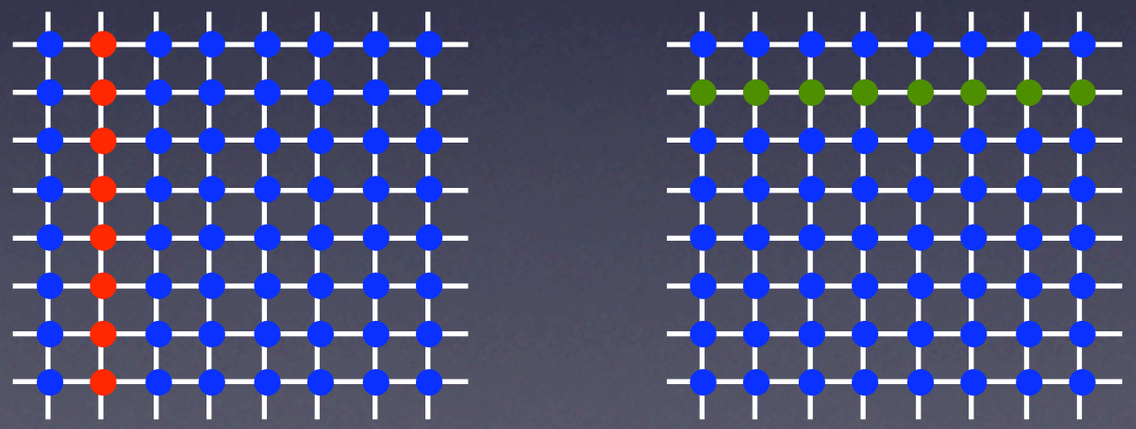
group generated by



abelian invariant subgroup:



logical qubit operators:

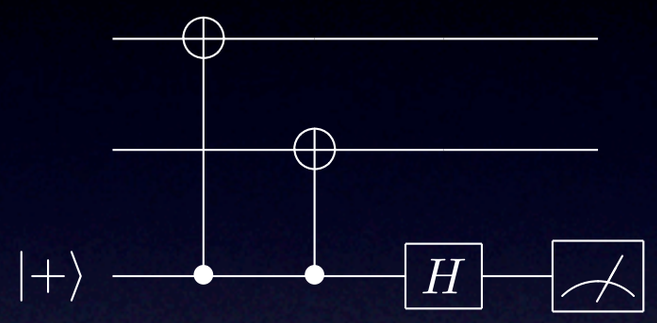


L^2 qubits
1 encoded qubit
corrects L/2 errors
L^2-2L-3 gauge qubits

Bacon, "Operator Quantum Error-correcting Subsystems for Self-correcting Quantum Memories" (2006)

Fault-Tolerance in Subsystem Code

gauge degrees of freedom allow fault-tolerant syndrome measurement two qubits at a time:



Leads to proven threshold for fault-tolerant quantum computing which is rather good (best without significant overhead:

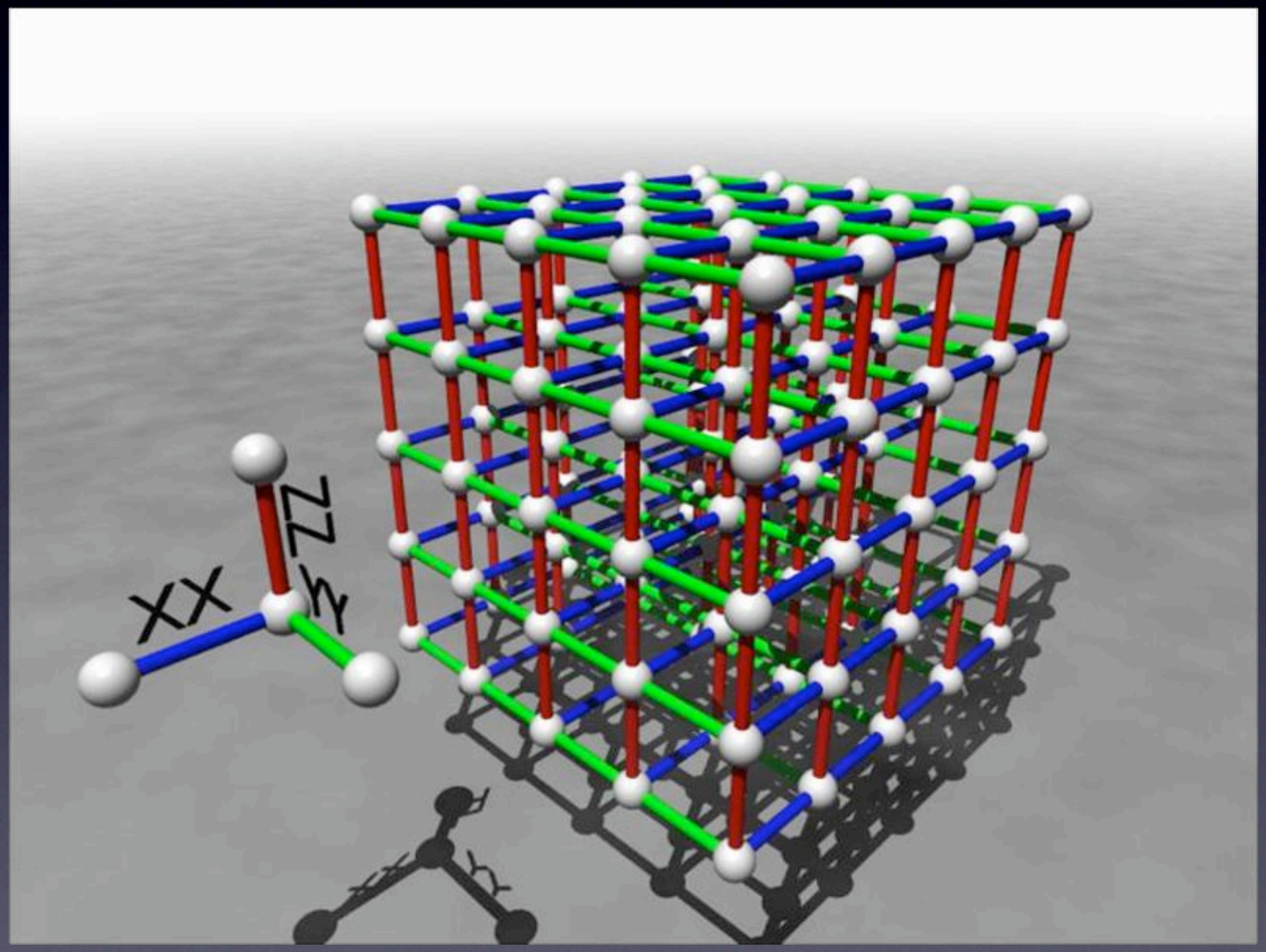
$$p^* > 2 \times 10^{-4}$$

Aliferis, Cross "Subsystem fault tolerance with the Bacon-Shor code" (2007)

And to good local architectures (see also Sandia group)

Spedalieri, Roychowdhury "Latency in local, two-dimensional, fault-tolerant quantum computing"

Possibly Self-Correcting



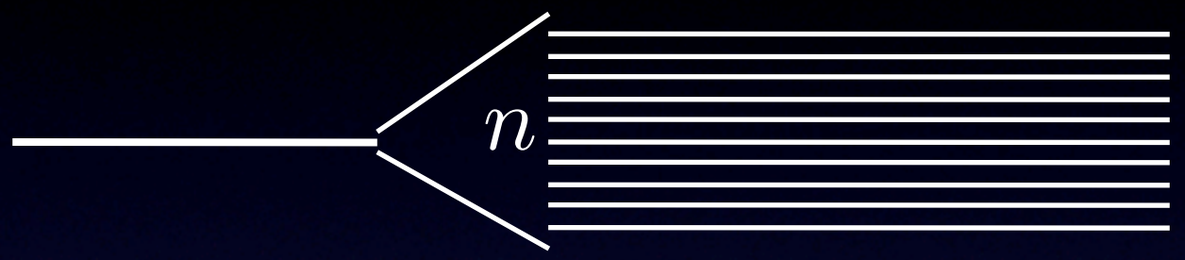
Bacon, "Operator Quantum Error-correcting Subsystems for Self-correcting Quantum Memories" (2006)

Concatenation

failure probability

p

Concatenation



failure probability

p

cp^2

Concatenation



failure probability

p cp^2 $c(cp^2)^2$

Concatenation



failure probability

p $\quad cp^2 \quad c(cp^2)^2$

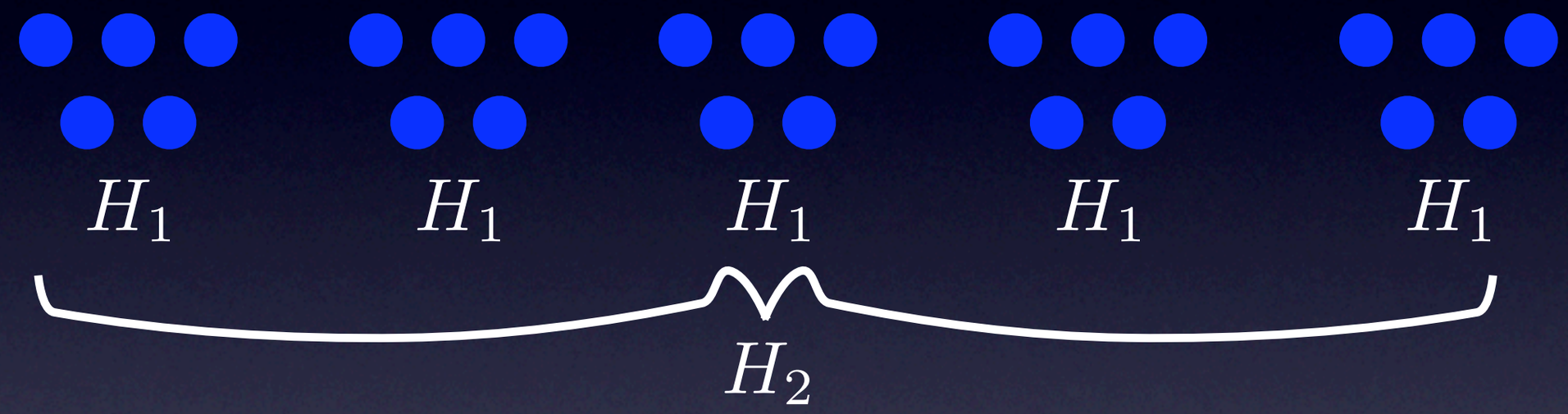
$$\frac{p_{\text{fail}}}{p_{\text{th}}} = \left(\frac{p}{p_{\text{th}}}\right)^{2^k} \qquad p_{\text{th}} = \frac{1}{c} \qquad N_{\text{qubits}} = n^k$$

$$N_{\text{qubits}} = \left[\log_{\frac{p}{p_{\text{th}}}} \left(\frac{p_{\text{th}}}{p_{\text{fail}}}\right)\right]^{\log_2 n} = O\left(\text{polylog}\ p_{\text{fail}}^{-1}\right)$$

below threshold, failure suppressed exponentially in number of qubits

Concatenated Hamiltonians

[[n, l, d]] base code



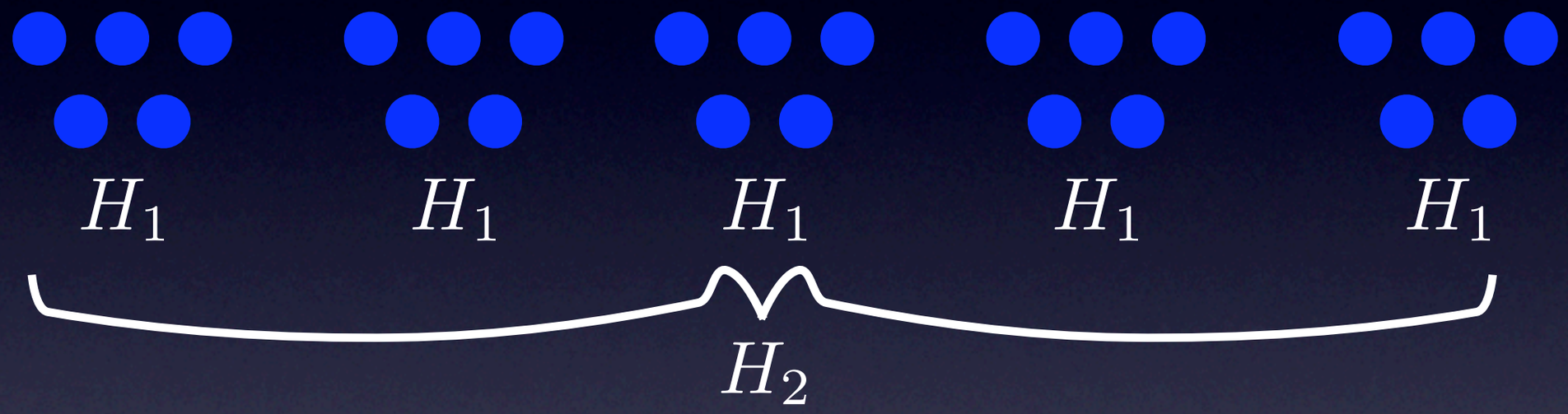
r levels of concatenation:

- n^r qubits
- Hamiltonian is sum of n^r-1 terms
- interactions are weight c^r operators

Bacon, "Quantum Concatenated Code Hamiltonians" (2008)

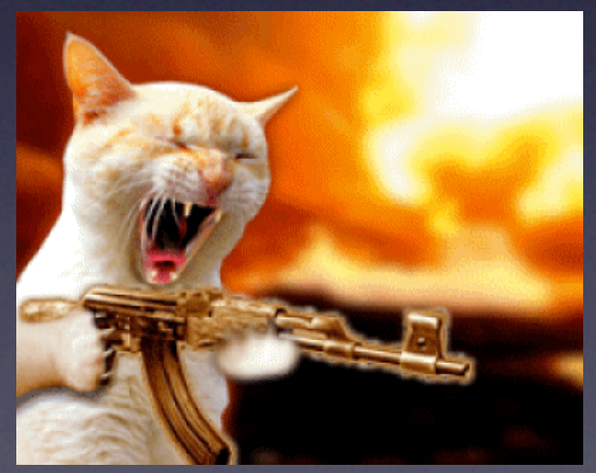
Concatenated Hamiltonians

[[n, 1, d]] base code



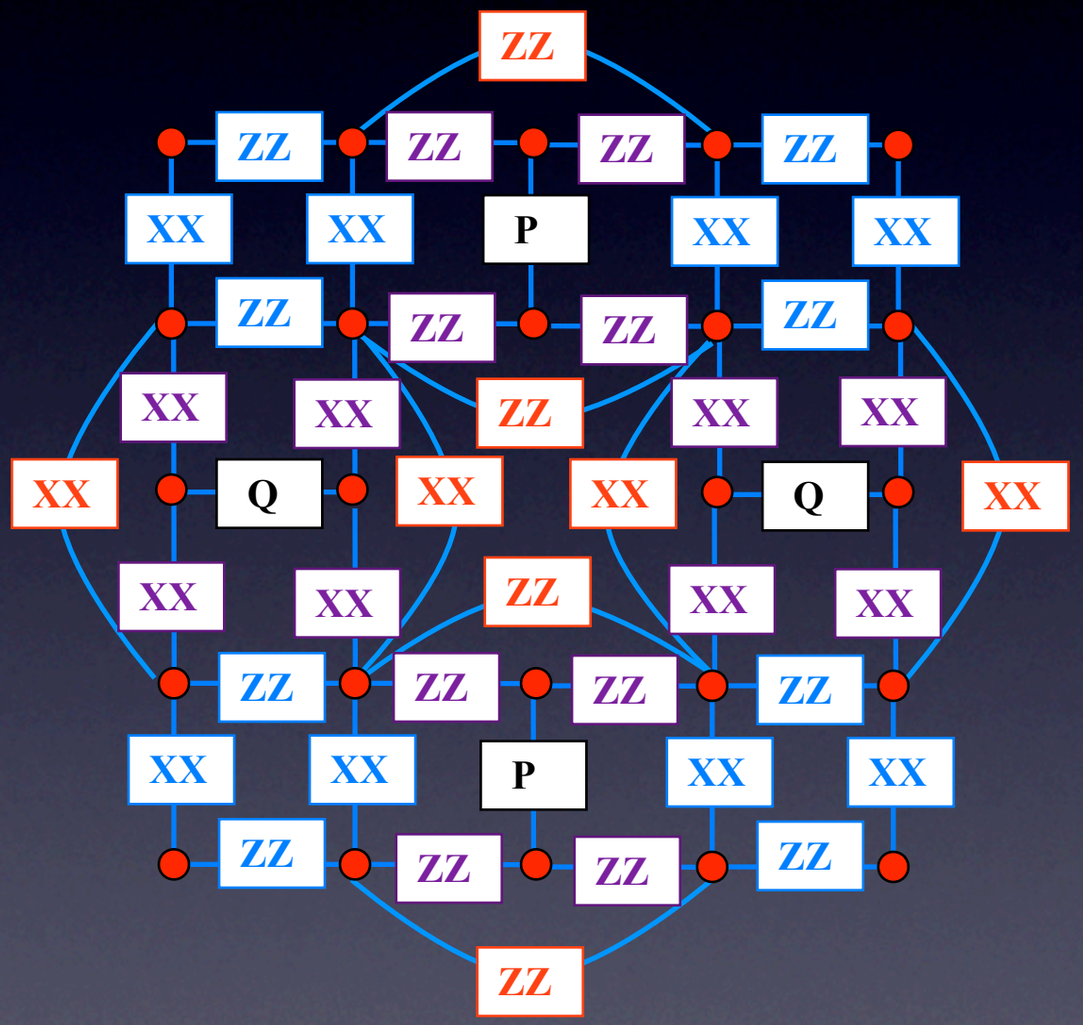
r levels of concatenation:

- n^r qubits
- Hamiltonian is sum of n^r-1 terms
- interactions are weight c^r operators

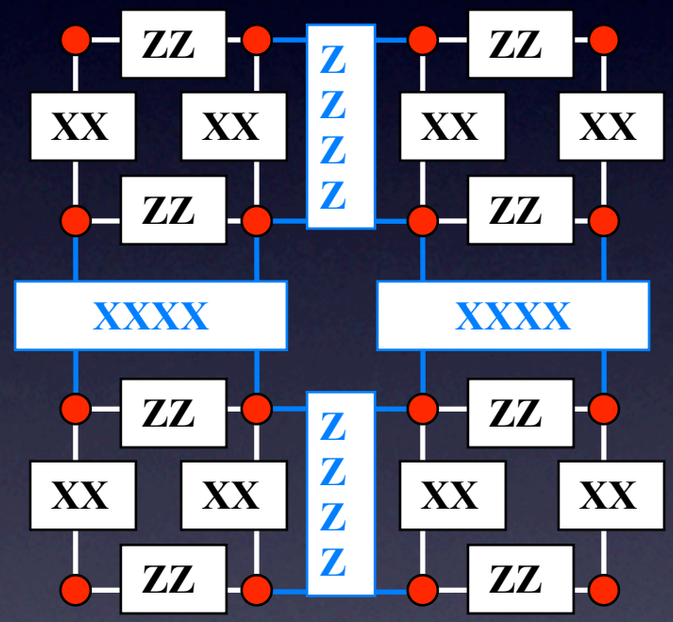


Bacon, "Quantum Concatenated Code Hamiltonians" (2008)

Self-Correcting 24 qubit system of distance 4



≅



concatenated [[16,1,4]] code

The Race

Fault-tolerant quantum computing justifies the model of quantum computing

But do we really need it, or can physics based methods be used?

Quick! The computer engineers are stealing all the physicists' ideas*

*it wouldn't be the first time that has happened...

Quantum Computing Theory Group

Graduate Students

Gregory Crosswhite (Physics)
Alice Neels (CSE)

Masters Students

Roger Wolfson (Physics)

Undergraduates

Elizabeth Muhm (CSE/Math)
Alper Sarikaya (CSE/Chem)
William Johnson (CSE)

funded by

ARO NSA NSF DARPA